Certified Commitment Horizons in Dynamic Lot Sizing

Exact Prefix Stability, Optimal-Face Witnesses and One-Step Forecast Calibration

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Assurance status. This candidate release separates the abstract forced-source path model from its classical lot-sizing interpretation. The latter is invoked only under the proved active-horizon bridge: every admissible revision has positive first-period demand support, and every later setup cost is strictly positive. Every value labelled *certified* is computed or re-verified in exact rational arithmetic with zero tolerances; floating-point optimisation proposes candidates but does not decide certification, topology or infinity. The frozen package contains source, tests, raw audit rows, a separately written certificate checker, version-matched provenance, file hashes and a source-specific inspection receipt for Richter–Vörös (1989). It has not received journal peer review, proof-assistant verification, independent clean-room reproduction or prospective field validation.

Abstract

Repeated forecast revisions can change near-term production setups after an organisation has begun to act. We define the *strong prefix-stability radius*: the weighted, floor-truncated L_∞ distance from a nominal demand forecast to the nearest revision at which some optimal replenishment path can leave a protected setup prefix. The mathematical core is a forced-source Wagner–Whitin path graph. A new semantic-bridge theorem proves that this graph is exactly the classical active-horizon uncapacitated lot-sizing model, at the level of optimal setup schedules, when the first-period demand floor is positive and all later setup costs are strictly positive. Inputs outside that domain remain valid graph instances but are labelled graph-only; this prevents leading-zero demand or costless empty setups from masquerading as operational instability.

Ordered reduced slopes make every path cost affine in demand. We derive an exact floor-aware distance to a tie with a fixed path, a prefix-deviation directed acyclic graph, and a polynomial frozen-comparator lower certificate. The certificate is exact for a complete path and can be strict for a partial prefix. A compact inverse linear programme supplies a certified upper witness, so coincident bounds close the strong radius without enumeration. We then prove three structural results. The longest prefix common to every nominal optimum determines whether the radius is positive. The exact prefix radius is the minimum of at most k one-protected-arc radii on suffix sub-instances. At every finite nearest destabilisation, the optimal face contains a prefix-retaining and a prefix-leaving adjacent path pair; those paths differ on one contiguous divergence–remergence detour. This compresses every boundary witness to a single structural exchange.

Exact escalation remains finite but worst-case exponential in the released enumeration; the demand-induced one-arc complexity classification remains open. A polynomial-size production-balance mixed-integer formulation avoids exponential front-loading while preserving unresolved solver states. One-step split-conformal calibration controls the next forecast revision only under exchangeability, and a deterministic rolling bound prices structural commitment under quantity recourse. Exact finite audits include the semantic bridge, optimal-face edge witnesses, fixed-path certificates, disjunctive closures, compact-model equivalence and rolling cases. An illustrative demand-series exercise uses reconstructed forecast vintages and stylised costs; it demonstrates the interface, not operational benefit. A page-specific negative reading of Richter–Vörös (1989) found fixed-solution scalar-parametric stability but no protected-prefix object; this supports the stated contribution boundary without asserting exhaustive priority.

Keywords: dynamic lot sizing; Wagner–Whitin; nervousness; commitment horizon; stability radius; inverse optimisation; shortest path; conformal calibration; rolling horizon.

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1 The decision problem

A fixed calendar fence answers a scheduling convention, not a robustness question. The relevant decision is instance-specific:

> How far must the current demand forecast move before an optimal schedule may change any protected setup epoch or replenishment-cycle boundary?

Semantic contract. The graph theorems use a *forced-source path model*: every complete path begins at period 1, and graph paths remain distinct even when they differ only by a zero-quantity, zero-cost setup node. This is an exact combinatorial model in its own right. We invoke the ordinary classical lot-sizing interpretation only in *classical active-horizon mode*, which requires

$$\ell_1 > 0, \qquad K_i > 0 \quad (i = 2, \dots, T). \tag{1}$$

Theorem 2.2 proves that these conditions make the forced source productive for every admissible revision and remove encoding-only ties from empty later setups. Every software record declares one of these two modes. Unless a result is explicitly called a classical corollary, its universal mathematical statement is at graph level.

Periods are $1, \dots, T$. A complete path

$$P = (v_0 = 1 < v_1 < \cdots < v_m = T + 1)$$

encodes replenishment cycles: arc (v_r, v_{r+1}) sets up in period v_r and covers demand through period $v_{r+1} - 1$. Fix a nominal forecast $\widehat{d} \in \mathbb{R}^T_+$, a selected nominal optimal path $P^\star$, and its first k arcs

$$\pi_k = (v_0, \dots, v_k).$$

Let

$$\mathcal{I}_k = \{P : P \text{ starts with } \pi_k\}, \qquad \mathcal{O}_k = \mathcal{P} \setminus \mathcal{I}_k,$$

and define the best prefix-retaining and prefix-leaving costs at revised demand d by

$$A_k(d) = \min_{P \in \mathcal{I}_k} C_P(d), \qquad B_k(d) = \min_{Q \in \mathcal{O}_k} C_Q(d).$$

A tie already permits a rational schedule change. Thus the prefix is compulsory for every optimum exactly when $B_k(d) > A_k(d)$.

Let $w \in \mathbb{R}^T_{++}$, let $\ell \in \mathbb{R}^T_+$ satisfy $\ell \le \widehat{d}$, and put

$$\|x\|_{\infty,w} = \max_t w_t |x_t|.$$

Weights set the coordinate scale; they are directionally symmetric. Direction-specific revision penalties would require a genuinely asymmetric gauge and are outside the present results. Floors are declared input semantics, not quantities inferred by the method: a floor may encode empirical or physical demand support, a contractual or planning minimum, or a deliberately conservative truncation. These interpretations produce the same mathematical lower bound but are not operationally interchangeable, and the provenance of an application must identify which is intended.

Definition 1.1 (Strong prefix-stability radius)**.** The weighted strong radius of the first k cycles is

$$\rho_{k,w}(\widehat{d}) = \inf \left\{ \left\| d - \widehat{d} \right\|_{\infty,w} : d \ge \ell,\ B_k(d) \le A_k(d) \right\}, \tag{2}$$

with the convention $\inf \varnothing = +\infty$. For $w = \mathbf{1}$ we write $\rho_{k,\infty}$.

The radius freezes structural setup epochs and cycle boundaries. It does not freeze quantities. Under the uncapacitated model, a retained cycle can adjust its order quantity to revised or realised demand.

Decision problem 1.2 (WW-Prefix-Radius$_\infty$)**. Input:** rational $K, q, \widehat{d}, \ell, w, R$, a declared semantic mode, a nominally optimal path $P^\star$, and $k \in \{1, \dots, m\}$. Classical active-horizon mode must satisfy (1); otherwise the instance is interpreted as a forced-source graph. Nominal optimality may be treated as a promise or checked by one shortest-path calculation.
Question: does some $d \geq \ell$ satisfy

$$\left\| d - \widehat{d} \right\|_{\infty,w} \leq R \quad \text{and} \quad B_k(d) \leq A_k(d)?$$

Equivalently, is $\rho_{k,w}(\widehat{d}) \leq R$?

Proposition 1.3 (Basic geometry)**.** *Assume $P^\star$ is nominally optimal.*

(a) For every $r < \rho_{k,w}(\widehat{d})$, every optimum at every admissible d with $\left\| d - \widehat{d} \right\|_{\infty,w} \leq r$ retains π_k.
(b) If the radius is finite, a nearest destabilising point exists. At a nearest point, $A_k(d) = B_k(d)$, except that $\rho_{k,w} = 0$ may already be a nominal tie; even then $A_k(\widehat{d}) = B_k(\widehat{d})$.
(c) $\rho_{k+1,w}(\widehat{d}) \leq \rho_{k,w}(\widehat{d})$ whenever both prefixes exist.
(d) The value depends only on the node prefix π_k, not on which nominal optimum containing that prefix supplied it.

Proof. Part (a) is the complement of the event in (2). For (b), intersect the admissible domain with a closed ball whose radius exceeds the infimum. This set is compact. Both A_k and B_k are minima of finitely many affine functions and hence continuous, so a convergent destabilising sequence attains the infimum. If $B_k < A_k$ at a nearest point, continuity along the segment from $\widehat{d}$ gives an earlier equality. For (c), $\mathcal{I}_{k+1} \subseteq \mathcal{I}_k$ and $\mathcal{O}_{k+1} \supseteq \mathcal{O}_k$. Part (d) follows because these two path classes are determined by π_k alone. □

2 Reduced cumulative-demand arcs

Let $K_i \geq 0$ be setup cost, p_i unit production or purchase cost, and $h_i \geq 0$ the cost of carrying one unit from i to $i + 1$. Demand is met without backlogging. Under the Wagner–Whitin no-speculative-motive condition

$$p_i + h_i \geq p_{i+1} \qquad (i = 1, \dots, T - 1), \tag{3}$$

an optimal plan has the zero-inventory-ordering structure used by the shortest-path model (Wagner and Whitin 1958).

Define

$$H_i = \sum_{s=1}^{i-1} h_s, \qquad q_i = p_i - H_i, \qquad D_1 = 0, \qquad D_j = \sum_{t=1}^{j-1} d_t.$$

Condition (3) is equivalent to

$$q_1 \geq q_2 \geq \cdots \geq q_T. \tag{4}$$

Proposition 2.1 (Reduced arc form)**.** *The physical cost of arc (i, j) equals*

$$\bar{c}_{ij}(d) + \sum_{t=i}^{j-1} H_t d_t, \qquad \bar{c}_{ij}(d) = K_i + q_i(D_j - D_i) = K_i + q_i \sum_{t=i}^{j-1} d_t. \tag{5}$$

Summed along any complete path, the local carrying terms assemble the path-independent quantity $\sum_{t=1}^{T} H_t d_t$. Therefore reduced and physical costs have the same optimal paths and the same path-to-path regret.

Table 1: Dependency and interpretation of the main results.

Result layer	Formal domain	Interpretation ceiling
Affine paths, tie distance, prefix graph, frozen certificate, exact bracket, disjunction, local decomposition and edge witness	Forced-source directed acyclic graph	Exact combinatorial statements for the declared graph semantics
Classical active-horizon corollary	Graph layer plus (1)	Optimal setup epochs and replenishment-cycle boundaries in the ordinary uncapacitated Wagner–Whitin model
One-step calibration	Deterministic radius plus exchangeable revision scores	Marginal coverage for the next forecast revision only
Rolling loss bound	Retained setup structure plus uncapacitated quantity recourse	Deterministic structural-cost bound; no service or capacity guarantee

Proof. For $t \geq i$, the physical unit cost of serving period t from setup period i is $p_i + \sum_{s=i}^{t-1} h_s = q_i + H_t$, so the arc's physical cost is $\bar{c}_{ij}(d)$ plus its local term $\sum_{t=i}^{j-1} H_t d_t$. Every complete path assigns period t to exactly one cycle, so $H_t d_t$ appears exactly once in the path total independently of the path. □

Theorem 2.2 (Classical active-horizon bridge)**.** *Assume zero initial inventory, no backlogging, the ordered slopes* (4)*, and the bridge conditions* (1)*. For every admissible $d \geq \ell$:*

(a) every feasible physical plan must open a productive setup in period 1*;*
(b) no optimal setup pattern contains a later open setup whose cycle serves zero demand;
(c) optimal physical setup patterns and optimal complete paths are in cost-preserving bijection, after removal of the path-independent carrying term in theorem 2.1.

Consequently, the graph quantities $A_k(d)$, $B_k(d)$ and $\rho_{k,w}(\widehat{d})$ are exactly the corresponding classical setup-schedule quantities in active-horizon mode.

Proof. Because $d_1 \geq \ell_1 > 0$, zero initial inventory and no backlogging force positive production, and hence an open setup, in period 1. Fix any set of open setups. Assign each unit of period-t demand to the latest open setup no later than t. Since $q_1 \geq \cdots \geq q_T$, this reassignment cannot increase reduced variable cost and preserves feasibility. It therefore yields the zero-inventory path whose nodes are the open setup periods. If a later setup i serves no demand before the next open setup, deleting i changes no assignment or variable cost and reduces setup cost by $K_i > 0$; such a node cannot occur in an optimum. Conversely, every complete path defines a feasible physical plan by producing each cycle's demand at its tail. The two constructions preserve the setup pattern and reduced cost, and theorem 2.1 adds the same carrying term to every path. □

Remark 2.3 (What the bridge excludes)**.** *If d_1 may be zero, the ordinary model can wait for its first productive setup, whereas the forced-source graph still pays K_1. For example, $K = (100, 1)$, $q = (1, 0)$ and $d = (0, 1)$ have graph optimum* 101 *but ordinary physical optimum* 1*. If a later $K_i = 0$, inserting an empty setup can create a distinct co-optimal graph path without changing an operational schedule. Such instances remain admissible in forced-source graph mode, where those semantics are explicit; they are rejected by classical active-horizon mode.*

Proposition 2.4 (Monge arcs)**.** *For $i < i' < j < j'$ and non-negative demand,*

$$\bar{c}_{ij}(d) + \bar{c}_{i'j'}(d) \leq \bar{c}_{ij'}(d) + \bar{c}_{i'j}(d).$$

Proof. The left side minus the right side is $(q_{i'} - q_i)(D_{j'} - D_j) \leq 0$. □

For every path P, let $i_P(t)$ be the setup period whose cycle contains t. Then

$$C_P(d) = \kappa_P + \beta_P^\top d, \qquad \kappa_P = \sum_{i \in P \setminus \{T+1\}} K_i, \qquad \beta_{P,t} = q_{i_P(t)}. \tag{6}$$

Ordered slopes provide the first proof attack: they create additive affine geometry and the exact coefficient diameter used in section 13. They do not collapse the exact prefix problem to one shortest path, because the minimiser of the inside envelope A_k can switch before an outside path wins.

Proposition 2.5 (Membership in NP)**.** *The rational decision problem WW-Prefix-Radius$_\infty$ belongs to NP: a yes-instance with input encoding length L has a certificate of length* poly(L) *verifiable in* poly(L) *exact rational arithmetic.*

Proof. A yes-certificate is a pair (Q, d): an outside path (at most $T+1$ node labels) and a rational revised demand vector. The verifier checks, in exact rational arithmetic, that Q is a complete path not retaining π_k; that $d \geq \ell$ and $w_t|d_t - \widehat{d}_t| \leq R$ for every t; and that $C_Q(d) \leq C_{\pi_k}(d) + V_s(d) = A_k(d)$, computing $V_s(d)$ by the $O(T^2)$ suffix recursion. Every intermediate value is a sum of at most $T+1$ products of input rationals with d-coordinates, so all numerators and denominators stay within poly(L) bits and the verification runs in polynomial time; no potentials need be supplied, since the verifier recomputes the inside envelope itself.

For the certificate-size bound, fix any $Q \in O_k$ witnessing the event. The feasible set $\{(d,u) : d \geq \ell,\ \left\|d - \widehat{d}\right\|_{\infty,w} \leq R,\ u_s = 0,\ u_j - u_i \leq \bar{c}_{ij}(d)\ (s \leq i < j \leq T+1),\ C_Q(d) \leq C_{\pi_k}(d) + u_{T+1}\}$ is a rational polyhedron with $O(T^2)$ rows whose coefficients are input data, and it is non-empty exactly when d can be chosen destabilising for Q (theorem 10.1): the potentials can always be raised to $u_{T+1} = V_s(d)$. A non-empty rational polyhedron contains a point of encoding length polynomial in the facet encoding length — a basic feasible solution whose entries are quotients of subdeterminants of the constraint data (Schrijver 1986, Cor. 10.2) — and projecting it to d gives the required poly(L)-bit witness. If nominal optimality of $P^\star$ is not promised, the verifier first checks it by one exact shortest path. The release ships this verifier as an executable function, exercised against certified witnesses and corrupted certificates in the test suite. □

No hardness result or unrestricted polynomial exact algorithm is claimed.

3 Corpus boundary and contribution boundary

Classical planning-horizon work asks how much future information is needed before early decisions stabilise (Eppen et al. 1969; Lundin and Morton 1975; Lee and Denardo 1986; Bes and Sethi 1988; Chand and Sethi 1990; Federgruen and Tzur 1994; Federgruen and Tzur 1995; Dawande et al. 2006; Dawande et al. 2007). Robust and stochastic lot-sizing work instead chooses plans under uncertainty in costs, demand evolution or lead times (Hanafizadeh et al. 2019; Forel and Grunow 2023; Weijers et al. 2025; Guillaume et al. 2026; Li and Romero Morales 2017). Those models optimise against an uncertainty description or forecast process. The present object conditions on one current finite forecast and asks the inverse local question: the demand-space distance to loss of compulsory-prefix optimality.

Interval-demand minimax-regret uncapacitated lot sizing is a closer algorithmic comparator. Zhang (2011) gives a polynomial dynamic programme for a two-stage model, and Li and Romero Morales (2013) give an $O(T^5)$ algorithm when demand alone is uncertain. Their decision order is

$$\min_y \max_{d \in \mathcal{U}} \big(F(y,d) - F^\star(d)\big) :$$

one setup vector y is selected before the interval-demand scenario, then evaluated by worst-case regret against a fully adaptive optimum. WW-Prefix-Radius$_\infty$ instead varies the radius of the uncertainty

box and asks whether there exists a nearest scenario at which the two *adaptive restricted envelopes* cross, $B_k(d) \le A_k(d)$. It therefore has different quantifiers, a protected optimal face rather than one fixed first-stage vector, and an inverse nearest-boundary objective rather than minimax regret. The polynomial minimax-regret algorithms do not by themselves classify WW-Prefix-Radius$_\infty$, but any eventual complexity proof must account for their regeneration-interval dynamic programmes.

The contribution boundary rests on a targeted search frozen on 18 August 2026, not on an exhaustive systematic review. Full text was inspected for Hoesel and Wagelmans (1993), the working-paper version of Chakravarti and Wagelmans (1998), Weijers et al. (2025), Li and Romero Morales (2013), Egri et al. (2014), and a seven-page print scan of Richter and Vörös (1989a). The supplied Pages bundle has SHA-256 `bd5ee5a82e399e11572b44d2bf334672be6bc6b47499271e636cf3a8f04aa255`; the copyrighted page images are not redistributed. A source-specific inspection receipt records pages 67–73 and the page-by-page comparison. The inspected literature PDFs are not redistributed; source URLs and content hashes are retained in the provenance and third-party notices.

Richter–Vörös parameterise the stationary-cost dynamic lot-sizing model in three ways: one scalar cost direction $(s + a\hat{s}, h + a\hat{h})$, one scalar demand direction $\widetilde{d}_t = d_t + b\widehat{d}_t$, and the joint two-parameter case. They derive intervals or a planar region within which one fixed regeneration-point solution remains optimal, then give a $T = 3$ example and IBM-XT timings through $T = 15$. The inspected source contains no protected-prefix class, no moving best prefix-retaining continuation, no compulsory-optimality tie event, no simultaneous T-dimensional weighted L_∞ demand ball, no coordinate floors and no complexity classification. No theorem-level collision was found *in this source*. We therefore treat it as the direct fixed-solution scalar-parametric antecedent. The negative reading supports the candidate contribution boundary but does not establish exhaustive priority. The same programme's adjacent papers (Richter 1987; Richter and Vörös 1989b) and Richter and Weber (1994) remain less completely inspected. The release records search strings, access states, exact hashes and missing full texts.

Partial inverse and partial anti-inverse optimisation. The first review of this manuscript omitted the body of work closest in spirit to its decision event. In *partial inverse* combinatorial optimisation, parameters are modified at minimum cost so that some optimum *contains* prescribed elements; in *partial anti-inverse* problems, so that some optimum *avoids* forbidden elements (Gassner 2009; Ley and Merkert 2025). A direct lot-sizing precedent is Egri et al. (2014): observed demand–optimal-lot-size pairs are used to infer setup, holding and backlog cost ratios through an inverse shortest-path linear programme. That problem rationalises observed complete decisions by changing latent costs; it does not perturb demand to find the nearest crossing of a protected optimal face. The present event—an optimum leaving a protected prefix—is of the anti-inverse family: destabilisation occurs exactly when some optimal path avoids completing the prefix pattern. Two hardness results calibrate expectations. Gassner (2009) proves that partial anti-inverse shortest paths are strongly NP-hard under both L_1 and L_∞ cost modifications. Ley and Merkert (2025) prove that partial inverse shortest paths on a directed acyclic graph are NP-complete even when weights may only be increased and only one arc is prescribed.

The mapping to WW-Prefix-Radius$_\infty$ is, however, not an equivalence, and the differences are load-bearing. First, the perturbed object here is the *demand vector*, which moves every arc cost through the correlated affine map $\bar{c}_{ij}(d) = K_i + q_i(D_j - D_i)$; the cited hardness constructions modify arc weights coordinatewise and independently. Second, the perturbation geometry is a floor-truncated weighted L_∞ ball on T demand coordinates, not a norm on $O(T^2)$ arc weights. Third, ordered slopes impose Monge structure absent from the generic constructions. Fourth, the tie semantics are strong: any outside optimum defeats the compulsory prefix. Consequently, the generic NP-hardness results do *not* settle the complexity of WW-Prefix-Radius$_\infty$, and the polynomial frozen certificate of section 7 does not contradict them: it bounds, rather than solves, the exact partial anti-inverse quantity. The correct statement of the open problem is therefore narrower than in the previous version: the exact complexity of WW-Prefix-Radius$_\infty$ is open *only for the demand-induced, ordered-slope Wagner–Whitin restriction*; the unrestricted arc-weight analogues are already hard. Table 2 records the comparison.

Table 2: Position of WW-PREFIX-RADIUS$_\infty$ within inverse shortest-path variants.

Dimension	WW-PREFIX-RADIUS$_\infty$ (this paper)	Complete inverse (Ahuja and Orlin 2001)	Partial inverse (Ley and Merkert 2025)	Partial anti-inverse (Gassner 2009)
Target object	prefix-leaving face of optima	one prescribed complete solution	optimum containing required elements	optimum avoiding forbidden elements
Perturbed data	demand vector (T coordinates)	arc costs	arc weights, increase only	arc weights
Induced dependence	correlated across all arcs via $q_i(D_j - D_i)$	coordinatewise	coordinatewise	coordinatewise
Norm	weighted L_∞ with floors	L_1, L_∞	various	L_1, L_∞
Graph structure	Wagner–Whitin DAG, Monge, ordered slopes	general	DAG suffices for hardness	general
Known complexity	restricted exact case open; polynomial certified bracket	often polynomial	NP-complete (one required arc)	strongly NP-hard

We therefore make no novelty claim for generic stability radii or the full-path special case. The candidate mathematical core is narrower and inspectable: distance to loss of compulsory optimality of a prefix face whose best continuation can change; a proved graph-to-classical semantic bridge; the prefix-deviation certificate and bracket; the local-arc decomposition; and the single-detour edge witness at every finite boundary. The statistical and rolling results integrate that geometry with explicitly bounded guarantees.

4 Nominal degeneracy and a canonical protected prefix

A solver may return any member of a degenerate nominal optimal face. A commitment rule should not mistake arbitrary tie-breaking for robustness.

Let $\mathcal{P}^\star(\widehat{d})$ be the nominal optimal path set. Construct the *optimal-arc graph*: arc (i, j) is retained when a shortest path from 1 to i, followed by (i, j) and a shortest suffix from j, has total value equal to the nominal optimum. Forward and backward dynamic programmes construct this graph in $O(T^2)$ time.

Theorem 4.1 (Common optimal prefix)**.** *Starting at node 1, follow the unique outgoing optimal arc for as long as one exists. The resulting node sequence is the longest prefix shared by every path in $\mathcal{P}^\star(\widehat{d})$. It is computable in $O(T^2)$ arithmetic operations.*

Proof. Every nominal optimum is a source–sink path in the optimal-arc graph, and every source–sink path in that graph is nominally optimal. If the current node has one outgoing optimal arc, every optimal path must take it. If it has two or more, optimal paths diverge immediately, so no longer common prefix exists. Forward and backward shortest-path values inspect all $O(T^2)$ arcs. □

Theorem 4.2 (Positive-radius criterion)**.** *Fix a prefix π_k contained in at least one nominal optimum.*

(a) If no complete path leaves π_k, then $\rho_{k,w} = +\infty$.
(b) Otherwise $\rho_{k,w} > 0$ if and only if every nominal optimum contains π_k.
(c) If two selected nominal optima contain the same π_k, their exact strong radii coincide. Their frozen-comparator certificates need not coincide because those certificates retain the full selected path as comparator.

Proof. Part (a) is immediate. If a nominal optimum lies outside, then $A_k(\widehat{d}) = B_k(\widehat{d})$ and the radius is

zero. Conversely, if every nominal optimum is inside and an outside path exists, the finite outside family has a strictly positive nominal gap above $A_k(\widehat{d})$. Continuity of the finite affine envelopes preserves this gap in some neighbourhood, giving a positive radius. Part (c) follows from theorem 1.3(d); the final statement follows because the frozen half-spaces compare outside paths with the selected complete path. □

Operationally, the release reports the number of nominal optima (capped), the longest common optimal prefix, and a deterministic tie-break (fewest setups, then lexicographic nodes). It only describes a prefix as nominally robust when theorem 4.2 supports that statement.

5 Exact distance to a fixed path tie

Fix an outside path Q and compare it with a complete nominal path $P^\star$. Define

$$g_Q = C_Q(\widehat{d}) - C_{P^\star}(\widehat{d}) \geq 0, \qquad b_Q = \beta_Q - \beta_{P^\star}. \tag{7}$$

Let $s_t = \widehat{d}_t - \ell_t$ be downward slack and define

$$\Psi_b(r) = r \sum_{t:b_t<0} \frac{|b_t|}{w_t} + \sum_{t:b_t>0} b_t \min\left\{\frac{r}{w_t}, s_t\right\}. \tag{8}$$

Theorem 5.1 (Floor-aware fixed-path tie radius)**.** *The exact distance from $\widehat{d}$ to $\{d \geq \ell : C_Q(d) \leq C_{P^\star}(d)\}$ is*

$$r(Q, P^\star) = \inf\{r \geq 0 : \Psi_{b_Q}(r) \geq g_Q\}. \tag{9}$$

At radius r, a gap-minimising revision is

$$d_t(r) = \begin{cases} \max\{\ell_t, \widehat{d}_t - r/w_t\}, & b_{Q,t} > 0, \\ \widehat{d}_t + r/w_t, & b_{Q,t} < 0, \\ \widehat{d}_t, & b_{Q,t} = 0. \end{cases} \tag{10}$$

If $g_Q = 0$, the radius is zero. If $g_Q > 0$ and $\Psi_{b_Q}(r) < g_Q$ for every r, the radius is infinite.

Proof. At fixed r, minimising the affine gap $g_Q + b_Q^\top(d - \widehat{d})$ over the floor-truncated weighted box separates by coordinate. Positive coefficients push demand down until the floor binds; negative coefficients push demand up. The maximum gap reduction is (8). The half-space is reached exactly when that reduction covers g_Q. □

The breakpoints are $w_t(\widehat{d}_t - \ell_t)$ for positive coefficients. Sorting them and scanning the piecewise-linear function computes the radius and an explicit revised-demand witness.

6 The prefix-deviation graph

Let $P^\star = (v_0 = 1, \ldots, v_m = T + 1)$ and protect $\pi_k = (v_0, \ldots, v_k)$. Define a directed acyclic graph $\mathcal{G}_k = (V_k, E_k)$ as follows.

$$V_k = \{M_r : r = 0, \ldots, k-1\} \cup \{D_j : j = 2, \ldots, T+1\},$$
$$s = M_0, \qquad t = D_{T+1}.$$

Its labelled arcs are:

(i) $M_r \to M_{r+1}$ labelled (v_r, v_{r+1}) for $r = 0, \ldots, k-2$;

(ii) $M_r \to D_j$ labelled (v_r, j) for $r = 0, \dots, k-1$ and $j \in \{v_r + 1, \dots, T+1\} \setminus \{v_{r+1}\}$;
(iii) $D_i \to D_j$ labelled (i, j) for $2 \le i < j \le T+1$.

The expected kth prefix arc is deliberately omitted. A matching state records that the path has not yet diverged; after divergence the ordinary physical directed acyclic graph takes over.

Lemma 6.1 (Path bijection)**.** *Reading physical labels maps source–sink paths in $\mathcal{G}_k$ bijectively to O_k. The graph has $O(T)$ vertices and $O(T^2)$ arcs.*

Proof. Every outside physical path has a unique first arc at which it differs from π_k. It follows matching arcs before that point, takes the corresponding divergence arc, and then follows uniquely labelled ordinary arcs. Conversely every expanded path takes one first divergence before completing the protected prefix and therefore maps to an outside path. The size bounds follow directly from the construction. □

Let $Nx = h$, $x \ge 0$, be the unit source–sink flow equations with the sink row removed.

Lemma 6.2 (No non-negative circulation in a directed acyclic graph)**.** *If $\mathcal{G}_k$ is acyclic, $Ny = 0$ and $y \ge 0$, then $y = 0$.*

Proof. Assume some arc has positive flow and choose the earliest topological vertex that is the tail of a positive-flow arc. No positive-flow arc enters it, since such an arc would have an earlier positive-flow tail. Its outflow-minus-inflow equation is therefore strictly positive, contradicting $Ny = 0$. □

This elementary lemma supplies the missing strict-positivity step in the Charnes–Cooper proof below.

7 A polynomial frozen-comparator certificate

Define the frozen radius

$$\rho^{\mathrm{fr}}_{k,w}(\widehat{d}) = \min_{Q \in O_k} r(Q, P^\star). \tag{11}$$

It compares every outside path with the selected complete nominal path, rather than with the moving inside envelope.

Theorem 7.1 (Validity, monotonicity and full-path exactness)**.** *For every protected prefix,*

$$\rho^{\mathrm{fr}}_{k,w} \le \rho_{k,w}.$$

The frozen radii are non-increasing in k. If $k = m$, so the full path is protected, then $\rho^{\mathrm{fr}}_{m,w} = \rho_{m,w}$.

Proof. At a destabilising d, choose Q attaining $B_k(d)$. Then $C_Q(d) = B_k(d) \le A_k(d) \le C_{P^\star}(d)$, so d belongs to a frozen pairwise half-space. The distance to their union is no greater than the distance to the true destabilising set. Monotonicity follows because O_k expands with k. For a full path, $A_m = C_{P^\star}$. □

For each coordinate put $\delta_t = w_t(\widehat{d}_t - \ell_t)$. Remove repeated values, keep distinct positive breakpoints

$$0 = \delta_0 < \delta_1 < \cdots < \delta_L < \delta_{L+1} = +\infty.$$

Coordinates with zero downward slack are treated as saturated on the first interval.

For an expanded arc e labelled (i, j) and a period $t \in \{i, \dots, j-1\}$, set $c_{e,t} = q_i - \beta_{P^\star,t}$. On interval $[\delta_\ell, \delta_{\ell+1}]$, define

$$a^\ell_e = K_i + \sum_{t=i}^{j-1} \begin{cases} c_{e,t}\widehat{d}_t, & c_{e,t} < 0, \\ c_{e,t}\ell_t, & c_{e,t} > 0 \text{ and } \delta_t \le \delta_\ell, \\ c_{e,t}\widehat{d}_t, & c_{e,t} > 0 \text{ and } \delta_t > \delta_\ell, \\ 0, & c_{e,t} = 0, \end{cases} \tag{12}$$

$$b_e^\ell = \sum_{t=i}^{j-1} \begin{cases} -c_{e,t}/w_t, & c_{e,t} < 0, \\ c_{e,t}/w_t, & c_{e,t} > 0 \text{ and } \delta_t > \delta_\ell, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

For a unit path flow x, its worst frozen gap is

$$F_x(r) = (a^\ell)^\top x - \kappa_{P^\star} - r(b^\ell)^\top x. \tag{14}$$

Theorem 7.2 (Polynomial certificate)**.** *The exact rational value $\rho^{\mathrm{fr}}_{k,w}$ is computable by at most $T+1$ shortest-path calculations and polynomial-size linear programmes on $\mathcal{G}_k$.*

At the left endpoint of each interval, minimise $F_x(\delta_\ell)$ by shortest path. If the minimum is non-positive, the first crossing is δ_ℓ. Otherwise solve

$$\min_{x \in X_k :\, (b^\ell)^\top x > 0} \frac{(a^\ell)^\top x - \kappa_{P^\star}}{(b^\ell)^\top x}, \tag{15}$$

where $X_k = \{x \geq 0 : Nx = h\}$. Its Charnes–Cooper form is

$$\begin{aligned} \min_{y,\tau} \quad & (a^\ell)^\top y - \kappa_{P^\star}\tau \\ \text{s.t.} \quad & Ny = \tau h, \\ & (b^\ell)^\top y = 1, \\ & y \geq 0, \quad \tau \geq 0. \end{aligned} \tag{16}$$

If (16) is infeasible, no path has positive denominator on that interval. The first feasible optimum lying at or before the right endpoint is $\rho^{\mathrm{fr}}_{k,w}$. If the final interval has no crossing, the radius is infinite.

Proof. Between consecutive breakpoints, the set of positive coefficient differences saturated at the floor is fixed. The coordinatewise formula in theorem 5.1 therefore distributes additively over arcs as (14). If the minimum at the left endpoint is positive, a path crosses in the interval exactly when its denominator is positive and its ratio in (15) lies in the interval.

For a denominator-positive path flow, set $\tau = 1/(b^\ell)^\top x$ and $y = \tau x$ to obtain (16). Conversely, suppose (16) is feasible. If $\tau = 0$, then $Ny = 0$ and $y \geq 0$, so theorem 6.2 gives $y = 0$, contradicting $(b^\ell)^\top y = 1$. Hence $\tau > 0$ and $x = y/\tau \in X_k$ has positive denominator. Every feasible unit flow decomposes into source–sink path flows. Its ratio in (15) is a denominator-weighted average of the ratios of its denominator-positive path components, while zero-denominator components cannot improve a safe left endpoint. Thus some path attains a ratio no larger than the flow ratio; equivalently, an optimum may be chosen at a network-flow extreme point. Endpoint ties are assigned to the first interval encountered; repeated and zero breakpoints have already been collapsed. A zero denominator cannot cross within an interval whose left endpoint is safe. On the final interval, infeasibility of (16) means no remaining path can reduce its gap, so the radius is infinite.

There are at most $T+1$ intervals. The graph has $O(T)$ vertices and $O(T^2)$ arcs; each linear programme has $O(T^2)$ variables and constraints and rational encoding length polynomial in the input. This is a polynomial bit-complexity statement, not a strongly polynomial algorithm claim. □

An optimal extreme point recovers an outside path. Substituting the returned radius into (10) recovers a revised-demand witness.

Exact rational implementation and independently checkable certificates

Assurance boundary. Theorem 7.2 is a statement in exact rational arithmetic, and *so is the released implementation of every certified output*. Floating-point linear programming appears only as a search heuristic whose proposals (witness demands and dual mixtures) are re-verified exactly before they may influence any certified value or classification. Quantities that floating point computes but the exact layer has not verified are published under a separate `float_search` key labelled diagnostic. No tolerance parameter participates in any zero test, saturation decision, breakpoint comparison, tie classification or infinity claim.

The first review of this manuscript exhibited a valid two-period instance on which the version 2.0.0 floating-point implementation discarded a 10^{-9} coefficient difference under a 10^{-8} absolute tolerance, inferred an unreachable topology, promoted the frozen radius to infinity, and certified a proposed radius of 1.1 as safe although a destabilising revision existed at radius 1 in the reviewer's decimal arithmetic. The defect occurred before optimisation, so no post-hoc safety margin could repair it. Version 2.1.0 removes the failure class rather than the instance: binary floating-point inputs convert losslessly to rationals, exact decimal semantics are available through numeric strings in the input schema, and the certificate computation itself is exact.

The exact algorithm follows the interval structure of theorem 7.2 but replaces the Charnes–Cooper programme with a parametric shortest-path iteration that terminates finitely in exact arithmetic. Three observations make this work. First, the global minimal frozen gap $r \mapsto \min_{Q \in \mathcal{O}_k}\{g_Q - \Psi_{b_Q}(r)\}$ is continuous and non-increasing, so the crossing interval is found by *binary search over the sorted distinct breakpoints*, evaluating one exact shortest path per probe. Second, within an interval the minimal gap is a concave piecewise-linear function of r with non-positive slopes, so Newton–Dinkelbach iteration on the active piece descends monotonically and lands exactly on the first crossing after finitely many exact shortest paths; a flat active piece at a strictly positive value would contradict positivity at the interval's left endpoint, which rules out stalling. Third, in the final unbounded interval, a maximum-slope path computed by one shortest path either seeds the iteration or proves that every outside path has exactly zero remaining slope, in which case the residual minimal gap is itself a certificate of infinity. The implementation performs $O(\log T)$ exact shortest-path evaluations for the interval search plus a small number of Newton steps (at most the number of linear pieces; in the audit never more than six), each on the $O(T^2)$-arc deviation graph.

Every certified record carries evidence: the exposed outside path, an exact rational witness demand that ties the comparator exactly, its exact weighted distance, and—for infinite radii—either the structural absence of outside paths or the certified positive residual gap at full saturation. Internal cross-checks recompute the pairwise radius of the exposed path and fail loudly (never degrade) on any disagreement. A separately written checker, `scripts/check_certificate.py`, re-verifies published records from first principles without importing the package: it re-derives nominal optimality, replays witnesses, re-proves saturation infinity with its own dynamic programme, re-checks the triage logic, and for $T \leq 12$ re-derives frozen minimality by complete enumeration. This follows the certificate-checking paradigm used for exact mixed-integer programming (Cheung et al. 2017; Eifler and Gleixner 2021): trust is placed in a small independent verifier, not in the search code. The remaining gap to full formal assurance is stated in section 18: for $T > 12$ the checker confirms achievability, infinity and consistency but relies on the package's internally cross-checked parametric computation for minimality of the lower bound, and no proof assistant has verified either implementation (Huangfu and Hall 2018; Virtanen et al. 2020).

8 A polynomial lower–upper bracket

The frozen certificate is safe but may be conservative. It also exposes an outside path that can furnish a feasible upper bound.

For fixed $Q \in O_k$, let r^{inv}_Q be the minimum revision radius at which Q is no more expensive than the *best* prefix-retaining path. Section 10 gives its compact linear programme.

Theorem 8.1 (Polynomial bracket and exact closure)**.** *Assume $O_k \neq \varnothing$ and $\rho^{\mathrm{fr}}_{k,w} < \infty$, so an outside path attains the frozen radius; the empty and infinite cases are already settled by theorem 4.2(a) and theorem 7.1. Let $\widehat{Q}$ be an outside path attaining $\rho^{\mathrm{fr}}_{k,w}$ and define*

$$\rho^{\mathrm{lo}} = \rho^{\mathrm{fr}}_{k,w}, \qquad \rho^{\mathrm{up}} = r^{\mathrm{inv}}_{\widehat{Q}}.$$

Then

$$\rho^{\mathrm{lo}} \le \rho_{k,w} \le \rho^{\mathrm{up}}. \tag{17}$$

Moreover:

(a) if the frozen revised-demand witness for $\widehat{Q}$ also satisfies $C_{\widehat{Q}}(d) \le A_k(d)$, then $\rho_{k,w} = \rho^{\mathrm{lo}}$;
(b) if $\rho^{\mathrm{lo}} = \rho^{\mathrm{up}}$, then both equal the exact radius;
(c) if $\rho^{\mathrm{lo}} = +\infty$, then $\rho_{k,w} = +\infty$.

Both sides have polynomial-size formulations.

Proof. The lower inequality is theorem 7.1. The exact radius is $\min_{Q \in O_k} r^{\mathrm{inv}}_Q$, so it is at most the fixed value for $\widehat{Q}$. A closing lower witness is feasible for the exact destabilising set, giving the reverse inequality at ρ^{lo}. Coincident bounds close the interval. Finally, any exact destabilisation has some Q with $C_Q \le A_k \le C_{P^\star}$, so a finite exact radius would imply a finite frozen tie. □

At a proposed radius R, this gives a three-way polynomial triage:

$R <$ safe ρ^{lo}	: certified no destabilisation;
$R \ge \rho^{\mathrm{up}} < \infty$	: certified destabilising witness exists;
otherwise	: unresolved; exact escalation or no commitment.

The strict first inequality respects the tie boundary.

9 Why the frozen certificate can be strict

Proposition 9.1 (Positive-floor strict-gap instance)**.** *Let $T = 3$, $w = (1, 1, 1)$,*

$$q = (2, 1, 0), \quad K = (0, 3, 2), \quad \widehat{d} = (1, 5, 1), \quad \ell = (0.5, 0.5, 0.5).$$

The selected nominal optimum is $P^\star = (1, 2, 4)$ with cost 11. Protect arc $(1, 2)$. Then

$$\rho^{\mathrm{fr}}_{1,\infty} = 1.5, \qquad \rho_{1,\infty} = 2.$$

Proof. The four path costs are

$$\begin{aligned} (1,2,4): &\quad 2d_1 + d_2 + d_3 + 3, \\ (1,2,3,4): &\quad 2d_1 + d_2 + 5, \\ (1,3,4): &\quad 2d_1 + 2d_2 + 2, \\ (1,4): &\quad 2d_1 + 2d_2 + 2d_3. \end{aligned}$$

The first two paths are inside. The path $(1, 3, 4)$ has nominal gap 3 and coefficient difference $(0, 1, -1)$ relative to $P^\star$, so its frozen tie radius is $3/2$. At the witness $(1, 3.5, 2.5)$, however, the alternative inside path $(1, 2, 3, 4)$ costs 10.5, below both tied paths at 11.

For any admissible d with $\left\|d - \widehat{d}\right\|_\infty < 2$, $d_2 > 3$. If $d_3 \geq 2$, $(1,2,3,4)$ is best inside and its outside-minus-inside gap against $(1,3,4)$ is $d_2 - 3 > 0$. If $1 \leq d_3 < 2$, $P^\star$ is best inside and $(1,3,4)$ is best outside, with gap $d_2 - d_3 - 1 > 0$. If $d_3 < 1$, $P^\star$ is best inside and $(1,4)$ is best outside, with gap $d_2 + d_3 - 3 > 0$ because $d_3 \geq 0.5$. At $d = (1,3,3)$, distance 2, paths $(1,2,3,4)$ and $(1,3,4)$ tie at 10. Hence the exact radius is 2. □

This example isolates the moving-continuation effect. It also provides a regression test: the obvious continuous path-flow relaxation admits a half-integral solution at radius 1 even though the true event is impossible below 2.

10 Exact inverse formulations

10.1 One outside path

Let $s = v_k$ be the last node of π_k. Write its fixed cost as $C_{\pi_k}(d) = \kappa_{\pi_k} + \beta_{\pi_k}^\top d$. Introduce suffix potentials $u_s, \ldots, u_{T+1}$ (the symbol π is reserved for the protected prefix) satisfying

$$u_s = 0, \qquad u_j - u_i \leq K_i + q_i \sum_{t=i}^{j-1} d_t \quad (s \leq i < j \leq T+1). \tag{18}$$

The greatest feasible u_{T+1} is the suffix shortest-path value.

Theorem 10.1 (Fixed-competitor inverse linear programme)**.** *For fixed $Q \in O_k$, r_Q^{inv} is the optimum of*

$$\begin{aligned} \min_{d,r,u} \quad & r \\ \text{s.t.} \quad & d_t \geq \ell_t, \\ & -r/w_t \leq d_t - \widehat{d}_t \leq r/w_t, \\ & \pi \text{ satisfies (18)}, \\ & C_Q(d) \leq C_{\pi_k}(d) + u_{T+1}, \\ & r \geq 0. \end{aligned} \tag{19}$$

Consequently

$$\rho_{k,w} = \min_{Q \in O_k} r_Q^{\mathrm{inv}}. \tag{20}$$

Proof. For fixed d, shortest-path duality makes the maximum feasible terminal potential equal the best suffix value. The final inequality is therefore feasible exactly when Q is no more expensive than the best path retaining π_k. Minimising distance gives the fixed disjunct; minimising over outside paths gives the union. □

Equation (20) is an exact finite disjunction, not a polynomial algorithm: the number of disjuncts is exponential in T and candidate generation alone dominates observed runtimes (section 14). Sorting outside paths by the pairwise lower bound $r(Q, P^\star)$ gives an anytime algorithm: after solving some disjuncts, the incumbent is a feasible upper bound and the next unprocessed pairwise value is a lower bound. Stop exactly when they meet. On interruption, return both bounds, all statuses and any witness.

Certified escalation. The released escalation certifies both sides of (20) in exact arithmetic. Ordering and pruning use exact pairwise radii. Each processed disjunct's floating LP proposes a witness demand and dual values; the witness is rationalised and re-verified exactly against the moving inside continuation, and the duals are decomposed into an inside-path mixture λ whose separating power is re-verified through the exact halfspace walk: if the mixture $\sum_i \lambda_i C_{P_i}$ strictly dominates C_Q on the whole admissible box below some radius, that radius is a certified lower bound on r_Q^{inv}, because the inside minimum is dominated by every inside mixture. A one-point mixture at the binding inside path recovers r_Q^{inv} exactly in the nondegenerate case, and an auxiliary slack-minimisation probe supplies separating mixtures for disjuncts whose LP is infeasible, certifying $r_Q^{\mathrm{inv}} = +\infty$. The enumeration returns either a point-exact certified radius or a certified interval whose ends are both witness-backed; in the audit the interval, when present, had relative width below 10^{-12}, and its width is always published rather than rounded away.

10.2 A fixed-radius mixed-integer formulation

At radius R, put

$$L_t = \max\{\ell_t, \widehat{d}_t - R/w_t\}, \qquad U_t = \widehat{d}_t + R/w_t.$$

Place binary unit-flow variables z_e on $\mathcal{G}_k$. For each arc $e = (i, j)$ and $t = i, \dots, j-1$, introduce $u_{e,t} = d_t z_e$ and impose the exact binary McCormick hull

$$L_t z_e \le u_{e,t} \le U_t z_e, \tag{21}$$

$$d_t - U_t(1 - z_e) \le u_{e,t} \le d_t - L_t(1 - z_e). \tag{22}$$

The selected outside cost is

$$\sum_{e=(i,j)} K_i z_e + \sum_{e=(i,j)} q_i \sum_{t=i}^{j-1} u_{e,t}. \tag{23}$$

Theorem 10.2 (Fixed-radius mixed-integer model)**.** *Unit-flow constraints on $\mathcal{G}_k$, demand bounds $L \le d \le U$,* (21)–(22)*, suffix potentials* (18)*, and comparison of* (23) *with $C_{\pi_k}(d) + u_{T+1}$ form an exact mixed-integer formulation of WW-Prefix-Radius$_\infty$ at R. The direct formulation has $O(T^3)$ variables and constraints.*

Proof. Integer unit flow selects one outside path, and theorem 6.1 gives every outside path. With binary z_e, the McCormick inequalities are exact for $u_{e,t} = d_t z_e$ on the fixed box. Shortest-path duality makes the potential comparison exact. There are $O(T^2)$ arcs and $O(T^3)$ arc–period products. □

A feasible witness can be checked independently. A negative decision requires an optimal solver bound or a valid infeasibility certificate. Time limits with a feasible incumbent but no closing bound remain unresolved. The release rechecks a selected path through (19) to guard against mixed-integer tolerances near zero.

11 Enumeration complexity, arc decomposition and a compact exact model

This section characterises the exact escalation layer at three distinct levels: the size of the released disjunction, the structure of the decision problem, and a compact formulation that avoids arc–demand products. It gives the exact outside-path count, reduces the strong radius to single-arc events, states the rational witness bound used for NP membership, proves the single-detour optimal-face theorem, and derives a polynomial-size production-balance mixed-integer model. The mathematical arguments carry the universal claims. Finite exact-rational corpus tests exercise the corresponding implementations and identities; they do not constitute machine proofs of the theorems. Producer-side evidence is stratified in section 14.

11.1 Exact size and cost of the released disjunction

For a node $i \in \{1, \dots, T+1\}$, let $N(i)$ denote the number of paths from i to the terminal node $T+1$ in the complete lot-sizing graph, so $N(i) = 2^{\max\{T-i,0\}}$: a path from i chooses an arbitrary subset of the internal nodes $\{i+1, \dots, T\}$, and $N(T+1) = 1$.

Theorem 11.1 (Outside-path count)**.** *Let π_k end at $s = v_k$. Then*

$$|O_k| = N(1) - N(s) = 2^{T-1} - 2^{\max\{T-s,0\}},$$

and, partitioning by the first protected arc left, paths whose first divergence occurs at (v_r, v_{r+1}) number $N(v_r) - N(v_{r+1})$, which telescopes over $r < k$ to the same total. Every non-empty prefix has $s \geq 2$, hence

$$2^{T-2} \leq |O_k| < 2^{T-1}.$$

Proof. A complete path is determined by the subset of internal nodes $\{2, \dots, T\}$ it visits, giving $|\mathcal{P}| = N(1) = 2^{T-1}$. Retaining π_k fixes the inclusion of every protected node and the exclusion of every internal node below s that π_k skips, and leaves the subset of $\{s+1, \dots, T\}$ free, giving $N(s)$ inside paths (one when $s \in \{T, T+1\}$, where the free set is empty). Subtraction and the first-divergence partition are immediate; the bound uses $s \geq 2$ for any prefix with at least one arc. □

Corollary 11.2 (The released escalation is worst-case exponential)**.** *Both released escalation routines — the floating disjunction and its exact certified counterpart — materialise the full candidate family O_k, compute a pairwise lower bound per candidate and sort, before the first fixed-competitor programme is solved. Their candidate-generation stage therefore stores $\Theta(|O_k|)$ records and performs at least one pairwise tie-radius computation per record, hence $\Omega(|O_k|)$ work, which by theorem 11.1 is $\Omega(2^{T-2})$ for every non-empty prefix; ordering adds $O(|O_k| \log |O_k|)$ comparisons, and the complete escalation runs in $2^{O(T)}\,\mathrm{poly}(L)$ time on rational input of encoding length L. Lower-bound pruning acts only after generation and cannot reduce this stage. The count is available in closed form before any generation; this release discloses it and refuses escalations whose disjunction exceeds a caller-set cap.*

Remark 11.3 (The bound diagnoses the algorithm, not the problem)**.** *A full protected path has $|O_m| = 2^{T-1} - 1$, yet its exact radius is polynomial because the frozen certificate is exact there (theorem 7.1); the fixed-competitor programme (19) is polynomial for any single outside path. Exponential enumeration therefore says nothing about the complexity class of WW-Prefix-Radius$_\infty$; by theorem 11.8 below, the unresolved classification is confined to the one-protected-arc core.*

11.2 Local-tie decomposition of the strong radius

Fix the protected prefix $\pi_k = (v_0, \dots, v_k)$ of the nominal path. For revised demand d, let $V_i(d)$ be the optimal suffix value from node i (so $V_{T+1} = 0$), let

$$P_r(d) = \sum_{t<r} \bar{c}_{v_t v_{t+1}}(d)$$

be the cost of the prefix through v_r, and define at each protected arc

$$E_r(d) = \bar{c}_{v_r v_{r+1}}(d) + V_{v_{r+1}}(d), \qquad O_r(d) = \min_{j > v_r,\ j \neq v_{r+1}} \left\{ \bar{c}_{v_r j}(d) + V_j(d) \right\},$$

with $\min \varnothing = +\infty$ (the case $v_r = T$, $v_{r+1} = T+1$). E_r is the value of continuing along the protected Bellman arc and then optimally; O_r is the best value of leaving through any alternative arc. The protected arc at v_r is *compulsory* when $O_r > E_r$; the local tie $O_r \leq E_r$ says it has ceased to be so. Write also $W_r(d)$ for the best value from v_r among continuations that follow the remaining protected arcs $v_r \to \cdots \to v_k$ and then anything, so $W_k = V_{v_k}$, $W_r = \bar{c}_{v_r v_{r+1}} + W_{r+1}$, and $A_k(d) = P_r(d) + W_r(d)$ for every $r \leq k$.

Lemma 11.4 (First-arc split)**.** *$V_{v_r}(d) = \min\{E_r(d), O_r(d)\}$ for every $r < k$, and $E_r \le W_r$ pointwise.*

Proof. The Bellman recursion at v_r splits its outgoing arcs into the protected arc and the rest, giving the first identity. The second holds because $V_{v_{r+1}} \le W_{r+1}$: the protected continuation is one feasible suffix. □

Theorem 11.5 (Local-tie decomposition)**.** *For every admissible demand $d \ge \ell$,*

$$B_k(d) \le A_k(d) \quad \Longleftrightarrow \quad \exists\, r \in \{0, \dots, k-1\} : \; O_r(d) \le E_r(d).$$

All statements hold with weak inequalities, for zero demands, zero setup costs, and a prefix ending at the terminal node; no strictness or nondegeneracy is used.

Proof. ($\Leftarrow$) Suppose $O_r \le E_r$ at some $r < k$ and $O_r < \infty$. The path that follows the prefix to v_r, takes a minimising alternative arc (v_r, j), $j \ne v_{r+1}$, and continues with an optimal suffix from j is a complete path; its position $r+1$ differs from v_{r+1}, so it lies in O_k. Hence

$$B_k(d) \le P_r(d) + O_r(d) \le P_r(d) + E_r(d) \le P_r(d) + W_r(d) = A_k(d),$$

using theorem 11.4 for the middle inequality. If $O_r = +\infty$ the hypothesis forces $E_r = +\infty$, impossible in a finite graph.

($\Rightarrow$) By contraposition assume $O_r > E_r$ for every $r < k$. Then $V_{v_r} = E_r$ by theorem 11.4, and backward induction from $V_{v_k} = W_k$ gives $V_{v_r} = W_r$ for all $r \le k$: indeed if $V_{v_{r+1}} = W_{r+1}$ then $V_{v_r} = E_r = \bar{c}_{v_r v_{r+1}} + V_{v_{r+1}} = W_r$. Every outside path Q has a first divergence at some protected node v_r with $r < k$: it follows $v_0, \dots, v_r$ and then takes an arc (v_r, j) with $j \ne v_{r+1}$. Its cost obeys

$$\begin{aligned} C_Q(d) &\ge P_r(d) + \bar{c}_{v_r j}(d) + V_j(d) \\ &\ge P_r(d) + O_r(d) \\ &> P_r(d) + E_r(d) = P_r(d) + V_{v_r}(d) \\ &= P_r(d) + W_r(d) = A_k(d). \end{aligned}$$

Minimising over $Q \in O_k$ gives $B_k(d) > A_k(d)$. □

11.3 Optimal-face edge witnesses

Let X be the unit source–sink path polytope of the complete lot-sizing directed acyclic graph, and let χ^P denote the arc-incidence vector of path P. At demand d, the optimal face is

$$F(d) = \arg\min\{c(d)^\top x : x \in X\}.$$

The strong-radius boundary is therefore a face-crossing event, not merely a tie between two arbitrarily selected paths.

Theorem 11.6 (Single-detour optimal-face witness)**.** *If $A_k(d) = B_k(d)$, then $F(d)$ contains paths $P \in \mathcal{I}_k$ and $Q \in O_k$ whose incidence vectors are adjacent vertices of X. The paths coincide up to one divergence node a, follow two internally node-disjoint directed subpaths from a to one remergence node b, and coincide again from b to $T+1$. The detour crosses the protected-prefix boundary. Consequently, whenever $\rho_{k,w}(\widehat{d}) < \infty$, some nearest destabilising demand has a co-optimal prefix-retaining/prefix-leaving witness of this single-detour form.*

Proof. Equality $A_k(d) = B_k(d)$ supplies an optimal path in each class, so the optimal face contains vertices on both sides of the prefix partition. The graph of every polytope face is connected (Ziegler 1995, Ch. 3). Along a path in the one-skeleton of $F(d)$ from an inside vertex to an outside vertex, the first change of class is an edge $[\chi^P, \chi^Q]$ with $P \in \mathcal{I}_k$ and $Q \in O_k$.

It remains to identify the path exchange represented by an edge. If P and Q differed on two disjoint divergence–remergence intervals, swap only the first pair of subpaths to obtain two other source–sink paths R and S. Their incidence vectors satisfy

$$\chi^P + \chi^Q = \chi^R + \chi^S,$$

with $\{R, S\} \neq \{P, Q\}$. Hence the midpoint of $[\chi^P, \chi^Q]$ is also the midpoint of two other vertices, so P and Q cannot be adjacent. An adjacent pair therefore has exactly one divergence–remergence interval. Because the endpoints lie in different prefix classes, that interval changes at least one protected arc. The final assertion follows from theorem 1.3, which gives a nearest destabilising demand with $A_k = B_k$. □

Corollary 11.7 (Edge-face representation of the radius)**.** *For each single-detour crossing pair (P, Q) with $P \in \mathcal{I}_k$ and $Q \in \mathcal{O}_k$, let*

$$\mathcal{E}_{P,Q} = \{d \geq \ell : P \text{ and } Q \text{ are both globally optimal at } d\}.$$

Then

$$\rho_{k,w}(\widehat{d}) = \operatorname{dist}_{\infty,w}\left(\widehat{d}, \bigcup_{(P,Q)} \mathcal{E}_{P,Q}\right).$$

Thus every finite exact certificate admits a two-path, one-detour boundary witness. The representation does not by itself imply a polynomial algorithm: the family of edge faces can still be exponential.

Proof. Every edge face lies on the destabilising boundary, so the right-hand distance cannot be smaller than $\rho_{k,w}$. Conversely, theorem 11.6 places a single-detour edge face at a nearest finite destabilising point. If the radius is infinite, both sets are empty by the same theorem. □

Corollary 11.8 (Prefix-to-arc radius decomposition)**.** *Define the one-arc radius*

$$\rho_r = \inf\left\{\left\|d - \widehat{d}\right\|_{\infty,w} : d \geq \ell,\ O_r(d) \leq E_r(d)\right\}.$$

Then

$$\rho_{k,w}(\widehat{d}) = \min_{r<k} \rho_r,$$

every finite value is attained, and ρ_r depends only on the data of periods $v_r, \ldots, T$: it equals the strong radius, with one protected arc, of the suffix sub-instance whose nominal path starts with (v_r, v_{r+1}), with weights, floors and box coordinates restricted accordingly. Consequently WW-Prefix-Radius$_\infty$ with an arbitrary prefix reduces in polynomial time to at most k instances with a single protected arc, and since one arc is itself a special case, the two formulations are polynomially equivalent. The monotonicity $\rho_{k+1,w} \leq \rho_{k,w}$ of theorem 1.3 is recovered as minimisation over a larger arc set.

Proof. By theorem 11.5 the destabilising set $\{d \geq \ell : B_k \leq A_k\}$ is the union over $r < k$ of the local tie sets $\{d \geq \ell : O_r \leq E_r\}$, and the distance to a finite union is the minimum of the distances. Each local set is a finite union of polyhedra (one polyhedron per minimising alternative arc and suffix path, or the compact extended description with potentials as in (19)), hence closed, so finite distances are attained. E_r and O_r involve only arcs with tails at or after v_r, whose costs depend on $d_{v_r}, \ldots, d_T$ alone; coordinates before v_r are free and best left nominal. Interpreting the sub-problem on periods $v_r, \ldots, T$ with the shifted nominal path makes $O_r \leq E_r$ exactly the event $B_1 \leq A_1$ of that sub-instance. □

Example 11.9 (An interior link can fail first)**.** *Take the following exact decimal data with $T = 5$:*

$$K = (15.0,\ 44.3,\ 18.5,\ 31.3,\ 29.5), \qquad q = (0,\ -1.6,\ -3.4,\ -5.5,\ -7.2),$$
$$\widehat{d} = (86.8,\ 67.7,\ 43.1,\ 23.4,\ 97.9), \qquad \ell = (15.6,\ 10.9,\ 12.9,\ 3.5,\ 22.6),$$

$$w = (1.2,\ 1.8,\ 1.8,\ 1.1,\ 1.0).$$

The unique nominal optimum sets up in every period, $P^\star = (1,2,3,4,5,6)$; protect $k = 4$ arcs. The exact one-arc radii are

$$\rho_0 = \tfrac{28809}{400} = 72.0225, \qquad \rho_1 = \infty, \qquad \rho_2 = \tfrac{4906}{525} \approx 9.3448, \qquad \rho_3 = \infty,$$

so $\rho_{4,w} = \frac{4906}{525}$, attained at the third *protected arc: the certified enumeration over all $2^4 - 1$ outside paths returns the same rational. Local action stability is thus genuinely non-monotone along the prefix — protecting fewer arcs here changes nothing until the third link is included, and the first link is over seven times more robust than the prefix it heads.*

Remark 11.10 (The open core, restated)**.** *By theorem 11.8, the unresolved complexity classification of section 18 is exactly this: can a rational demand vector in the weighted floor-truncated box make one specified Wagner–Whitin outgoing arc cease to be uniquely optimal, under correlated costs $K_i + q_i(D_j - D_i)$, ordered slopes and the induced Monge structure? Generic partial inverse and anti-inverse hardness (Gassner 2009; Ley and Merkert 2025) modifies arc weights independently and does not settle this restriction in either direction.*

11.4 A compact production-balance exact model

The fixed-radius model of theorem 10.2 pays $O(T^3)$ arc–period products because path indicators multiply demand. Reformulating the declared forced-source graph over reduced production-balance variables removes every product. Under the bridge conditions (1), the same formulation has the classical active-horizon interpretation of theorem 2.2. Take variables $y_i \in \{0,1\}$ (setup), $x_i \geq 0$ (production), $I_i \geq 0$ (end-of-period inventory) for $i = 1,\dots,T$, demand d, and suffix potentials u_j for $j = s,\dots,T+1$ as in (18), and impose

$$I_i = I_{i-1} + x_i - d_i, \qquad I_0 = I_T = 0, \tag{24}$$

$$x_i \leq M y_i, \qquad M = \textstyle\sum_t U_t, \tag{25}$$

$$y_1 = 1, \tag{26}$$

$$\sum_{i<s,\ i\notin\pi_k} y_i \ + \sum_{v\in\pi_k\setminus\{1\},\ v\leq T} (1 - y_v) \ \geq \ 1, \tag{27}$$

with the demand box and floors of the mode in use. The plan cost is $\sum_i K_i y_i + \sum_i q_i x_i$; the fixed-radius decision minimises plan cost minus $C_{\pi_k}(d) + u_{T+1}$, and the minimum-radius variant instead couples plan cost $\leq C_{\pi_k}(d) + u_{T+1}$, adds $w_t|d_t - \widehat{d}_t| \leq r$ row pairs, and minimises r under an explicit cap. Constraint (26) encodes the forced source. In classical active-horizon mode it is also physically necessary because $d_1 \geq \ell_1 > 0$; outside that domain it is a graph-semantic constraint, not a claim about the unconstrained classical model. Constraint (27) states that the open-setup pattern differs from the protected one at or before s.

Lemma 11.11 (Conversion)**.** *Let (y,x,I,d) satisfy (24)–(26) with $d \geq 0$. Reassign every unit of period-t demand to the* latest *open setup at or before t. The result is the zero-inventory plan of the complete path whose nodes are exactly $\{i : y_i = 1\} \cup \{T+1\}$, its cost is at most $\sum_i K_i y_i + \sum_i q_i x_i$, and y is unchanged.*

Proof. By (24) and $I \geq 0$, demand in period t is covered by production in open periods $i \leq t$; (26) guarantees an open setup exists at or before every period. Moving one unit of period-t demand from an open setup i to the latest open setup $i' \in [i,t]$ changes its variable cost from q_i to $q_{i'} \leq q_i$ by the ordered slopes (4), so total variable cost does not increase; setup cost is unchanged because y is untouched. After reassignment each period is served entirely by its latest open predecessor, which is precisely the zero-inventory plan of the stated path. In forced-source graph mode, an open setup serving nothing

remains a node whose cycle carries zero quantity at cost $K_i + q_i \cdot 0$; this is an explicit graph decision. Interior zero demands are therefore permitted. In classical active-horizon mode, theorem 2.2 additionally proves that no optimal path can contain an empty later setup, because its strictly positive setup cost would be removable. □

Theorem 11.12 (Exactness of the compact forced-source model)**.** *Fix admissible $d \geq \ell$ in the declared forced-source graph. The minimum of $\sum_i K_i y_i + \sum_i q_i x_i$ over (24)–(27) equals $B_k(d)$. Consequently, with the potentials of (18): the fixed-radius decision model is an exact formulation of WW-Prefix-Radius$_\infty$ at radius R with T binary and $O(T)$ continuous variables and $O(T^2)$ constraints and no bilinear terms; and the minimum-radius model under a cap $R^{\max}$ is feasible exactly when $\rho_{k,w}(\widehat{d}) \leq R^{\max}$, in which case its optimal value is $\rho_{k,w}(\widehat{d})$; infeasibility certifies only $\rho_{k,w} > R^{\max}$, never an infinite radius.*

Proof. ($\geq$) Given a feasible plan, theorem 11.11 produces a complete path of no greater cost with the same y-pattern. Retention of π_k by that path would mean its node set contains every protected node up to s as consecutive initial nodes and no others below s; equivalently $y_v = 1$ for all $v \in \pi_k$, $v \leq T$, and $y_i = 0$ for every internal $i < s$ off the prefix — exactly the pattern excluded by (27). Hence the induced path lies in $\mathcal{O}_k$ and the plan cost is at least $B_k(d)$. ($\leq$) Any $Q \in \mathcal{O}_k$ defines the feasible plan that produces each cycle's demand at its setup node: its y-pattern violates prefix retention at its first divergence, satisfying (27) — an early divergence to $j < v_{r+1}$ opens a gap setup, a late one to $j > v_{r+1}$ closes the protected setup v_{r+1} (with $v_{r+1} \leq T$, since a divergence at $v_{r+1} = T + 1$ is impossible) — and its cost is $C_Q(d)$. The potential rows make $u_{T+1} \leq V_s(d)$ with equality achievable, exactly as in theorem 10.1, so comparing plan cost with $C_{\pi_k}(d) + u_{T+1}$ decides $B_k(d) \leq A_k(d)$; the radius rows are linear in (d, r). Attainment transfers from theorem 1.3. Under (1), theorem 2.2 transfers the result to classical active-horizon setup schedules; without those conditions the conclusion remains graph-level. □

Solver outputs remain floating point: as everywhere in this release, a finite compact-model radius or witness enters the certified record only after the exact layer re-verifies it (the binding-comparator tie of the disjunct certifier), and an infeasibility below the cap is solver-level evidence, never an infinity certificate. Section 14 reports a corpus equivalence gate — the compact model must reproduce every resolved exact $T \leq 18$ radius, witness and infinite-radius result of the existing audits before any scaling claim — and a preregistered benchmark against the released escalation.

12 One-step statistical calibration of forecast revisions

Three distinct objects must not be conflated, and each is controlled by a different result:

$$F^{\text{old}} = \widehat{d}, \qquad F^{\text{new}} = F^{\text{old}} + E_{n+1}, \qquad D^{\text{real}},$$

the current forecast, the next revised forecast, and realised demand respectively. The split-conformal result of this section is a *marginal, one-step* statement about the revision $F^{\text{new}} - F^{\text{old}}$ under exchangeability. Whether a reoptimisation at F^{new} can leave the protected prefix is then a *deterministic* consequence of the certified radius of sections 7 and 8. What commitment costs under D^{real} is controlled only by the deterministic rolling bound of section 13. No result in this paper calibrates realised-demand coverage, realised structural regret, service outcomes, or repeated rolling deployment.

The deterministic radius measures required movement, not the probability of movement. For aligned historical revisions E_i around base forecasts f_i, define the admissibility-aware score

$$S_i = \begin{cases} \|E_i\|_{\infty,w}, & f_i + E_i \geq \ell_i, \\ +\infty, & \text{otherwise.} \end{cases} \tag{28}$$

The infinite score prevents a calibration claim from silently extending beyond the domain in which the deterministic theorem holds.

For n scores and $\alpha \in (0, 1)$, let

$$j = \lceil (n+1)(1-\alpha) \rceil, \qquad q_{1-\alpha} = \begin{cases} S_{(j)}, & j \le n, \\ +\infty, & j = n+1. \end{cases}$$

Theorem 12.1 (One-step marginal revision tube)**.** *If $S_1, \dots, S_n, S_{n+1}$ are exchangeable, then*

$$\Pr\{S_{n+1} \le q_{1-\alpha}\} \ge 1 - \alpha,$$

where probability is marginal over the calibration sample and next revision.

Proof. The next score's rank among $n+1$ exchangeable scores is uniform under random tie-breaking and conservative for the non-randomised weak inequality. The probability that the rank exceeds j is at most α. □

Corollary 12.2 (One-decision calibrated commitment against the next forecast revision)**.** *For the current forecast $F^{\text{old}} = \widehat{d}$, let the next revised forecast be $F^{\text{new}} = F^{\text{old}} + E_{n+1}$ and use score (28). If*

$$q_{1-\alpha} < \rho^{\text{lo}}_{k,w}, \tag{29}$$

then the marginal probability that reoptimising at F^{new} may select an optimum leaving π_k is at most α.

Proof. On $S_{n+1} \le q_{1-\alpha}$, the revised forecast is admissible and lies strictly inside the deterministic lower radius. Theorem 12.1 bounds the complement. □

The corollary concerns the stability of the *plan under the next forecast*; it says nothing about the accuracy of either forecast for D^{real}, and a commitment that survives every forecast revision can still be costly if the forecasts themselves are poor. That risk is priced separately, and only in the worst-case pathwise sense, by theorem 13.3.

This is neither conditional coverage given the realised calibration set nor a simultaneous guarantee across repeated replans, products or adaptively selected groups. Weights, grouping, vintage alignment and missing-data rules must be fixed before the calibration scores used for coverage. For a prespecified finite set of M decisions, levels $\alpha_r > 0$ with $\sum_r \alpha_r \le \alpha$ and the union bound give family coverage at least $1 - \alpha$. Adaptive or drifting deployment needs a separate sequential method and its own assumptions; ordinary exchangeable split conformal does not inherit the guarantees of adaptive procedures (Barber et al. 2023; Gibbs and Candès 2021).

13 A rolling structural-loss theorem

At state a, let $V_a(d)$ be the optimal reduced suffix cost. A forecast $f \ge \ell$ chooses a prefix block $\pi : a \to b$ of an optimal suffix path. The block fixes setup arcs; quantities remain adjustable to realised demand. After executing the block, the planner replans at b. For realised $d \ge \ell$, define the Bellman residual

$$R_\pi(d) = C_\pi(d) + V_b(d) - V_a(d) \ge 0. \tag{30}$$

Lemma 13.1 (Telescoping residuals)**.** *If executed blocks concatenate from 1 to $T+1$, then*

$$C_R(d) - V_1(d) = \sum_r R_{\pi_r}(d).$$

Proof. Sum (30); intermediate value functions cancel and $V_{T+1} = 0$. □

For suffix paths from a, define

$$\Lambda_a = \max_{P,Q:a\to T+1} \sum_{t=a}^{T} \frac{|\beta_{P,t} - \beta_{Q,t}|}{w_t}.$$

Lemma 13.2 (Exact ordered-slope diameter)**.** *Under* (4)*,*

$$\Lambda_a = \Gamma_a := \sum_{t=a}^{T} \frac{q_a - q_t}{w_t}. \tag{31}$$

The direct suffix path and the path that sets up every period attain the maximum.

Proof. At period t, any suffix coefficient is q_i for some $i \in \{a, \dots, t\}$, so it lies in $[q_t, q_a]$. The stated two paths attain both endpoints coordinatewise. □

Theorem 13.3 (Certified dead-zone bound)**.** *Let ρ be the strong prefix radius of π around f over the admissible domain. For every realised $d \geq \ell$:*

(i) if $\rho = +\infty$, then $R_\pi(d) = 0$;
(ii) if $0 \leq \rho < \infty$, then

$$R_\pi(d) \leq \Gamma_a \left(\|d - f\|_{\infty,w,[a:T]} - \rho \right)_+ . \tag{32}$$

Replacing ρ by any certified lower radius preserves the inequality. Therefore a rolling policy obeys

$$C_R(d) - V_1(d) \leq \sum_r \Gamma_{a_r} \left(\|d - f^r\|_{\infty,w,[a_r:T]} - \rho_r^{\mathrm{lo}} \right)_+ . \tag{33}$$

Proof. If $\rho = +\infty$, no admissible demand makes an outside path beat the inside class, so the residual is zero. Now take finite ρ. The residual $A_\pi(d) - V_a(d)$ is zero throughout the closed radius ball. It is zero in the open ball by definition; if an outside path strictly beat the inside class at a boundary point, continuity along the segment from f would create an earlier tie. This also covers $\rho = 0$, since the forecast-selected inside path is nominally optimal.

For an inside optimiser P at an admissible point e with zero residual and an unrestricted optimiser Q at d,

$$R_\pi(d) \leq C_P(d) - C_Q(d) \leq (\beta_P - \beta_Q)^\top (d - e) \leq \Gamma_a \|d - e\|_{\infty,w,[a:T]} .$$

If d lies outside the ball and $\rho > 0$, choose the radial point on the suffix coordinates,

$$e = f + \frac{\rho}{\|d - f\|_{\infty,w,[a:T]}} (d - f),$$

with coordinates before a fixed and irrelevant to the suffix problem. The admissible domain is convex, so $e \geq \ell$, and $\|d - e\|_{\infty,w,[a:T]} = \|d - f\|_{\infty,w,[a:T]} - \rho$. For $\rho = 0$, take $e = f$. A smaller certified radius only enlarges the right side. Sum over blocks using theorem 13.1 and substitute theorem 13.2. □

The theorem prices structural setup commitments with uncapacitated quantity recourse. It does not cover fixed order quantities, capacity reservations, transport slots or service failures unless those decisions enter the feasible path model.

14 Computational audit

The release separates broad synthetic sampling, deliberately conditioned adversarial tests, solver scaling and the illustrative data case. All computations used seed 20260804. Raw rows include failures, infinities and status codes; conditioned strata are labelled so their frequencies cannot be read as prevalence estimates.

The 258 certificate instances comprise 162 broad draws and 96 instances deliberately conditioned to exhibit a strict frozen gap. Among positive partial-prefix broad cases, 5 of 72 were strict. All 96 conditioned target instances were strict by construction. The pooled partial lower/exact ratio had tenth percentile 0.641 and median numerically 1; the conditioned strict-gap regime had tenth percentile 0.618 and median 0.937. These numbers describe test coverage and conservatism, not an industrial distribution.

Exact certification audit

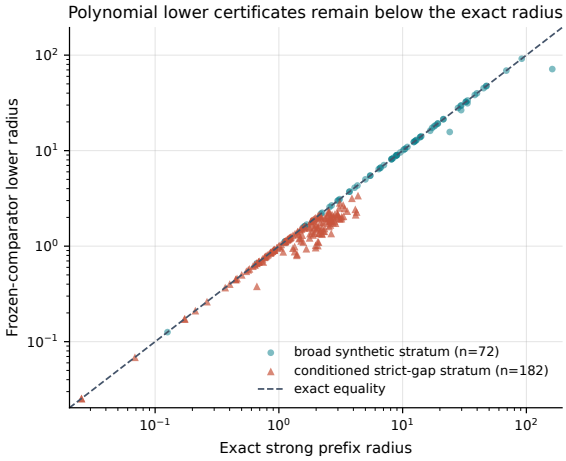
The floating-point comparisons above are retained as diagnostics of the search layer. Certification itself is audited by a separate deterministic script (`run_exact_certification_audit.py`) with three pre-specified strata; it fails loudly on any exact disagreement.

Agreement. Over 330 protected-prefix cases from 234 instances ($T \le 8$, broad, near-tie, floor-heavy and conditioned strict-gap regimes), the exact frozen certificate equalled complete enumeration of outside-path pairwise radii, in rational arithmetic, in every case; equality here is exact, not toleranced. The floating estimate never disagreed in finiteness on this stratum (no false infinite, no false finite) and its worst absolute error against the exact value was 4.5×10^{-13} — consistent with the earlier version's benign-range audits and with the reviewer's diagnosis that random small-instance agreement cannot detect the failure class, which lives outside those generators' numerical ranges. The certified escalation closed 305 of 330 cases point-exactly; the remaining 25 returned certified intervals whose worst relative width was 9.1×10^{-13}, each end witness-backed.

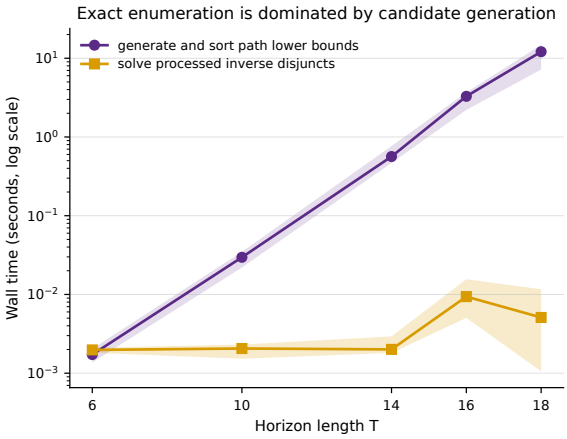
Adversarial. Twenty-seven deterministic families cover the review's counterexample in both input semantics, slope gaps from 10^{-3} down to 10^{-15}, uniform weights from 10^{-12} to 10^{12}, breakpoints one unit of least precision apart, and a 10^9 setup gap closed through a 10^{-12} slope difference. Under exact decimal semantics the counterexample certifies radius exactly 1; under binary semantics it certifies the exact radius of the instance actually supplied (≈ 1.0000000283); in both, a proposed radius of 1.1 is classified `certified_destabilising_witness_exists` and 0.9 is certified safe. A grid of fifteen exact power-of-two rescalings of cost and demand units (with q and w transformed accordingly) produced zero certificate or triage flips, asserted as a hard audit invariant; the invariance claim is scoped to rescalings that do not underflow or overflow in binary, and is unconditional when rescaling is performed in exact rational arithmetic.

Scaling. Table 6 reports certified runtimes at operational horizons on one commodity core. The interval binary search never needed more than 16 exact shortest-path evaluations, and Newton never more than 6 iterations, matching the $O(\log T)$-plus-pieces analysis; wall time is dominated by exact rational arithmetic on the $O(T^2)$-arc deviation graph. Minutes at $T = 365$ is compatible with batch planning cycles but not with interactive use; a scaled-integer kernel is the obvious engineering path and is left as disclosed future work.

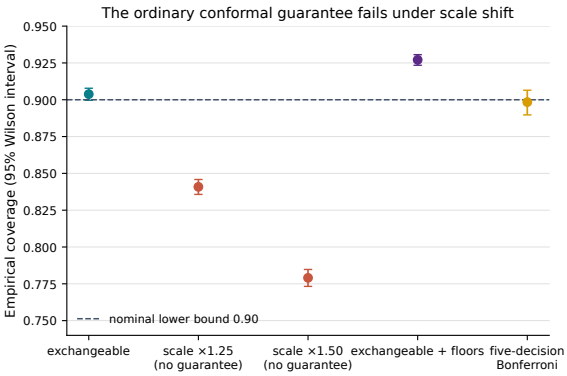
Cross-model adversarial review. Beyond the producer-side audit, the repair's soundness claims were submitted to an independent frontier model of a different vendor with instructions to refute them. It found no refutation of any certified classification as implemented and four caveats (input-domain preconditions, an exact-breakpoint crossing edge case, the inadmissibility of promoting mixture ties to strong witnesses, and the rescaling scope above), each verified against the code and locked with regression tests; the verbatim exchange and dispositions ship in the release documentation. This is additional adversarial scrutiny, not independent clean-room reproduction.



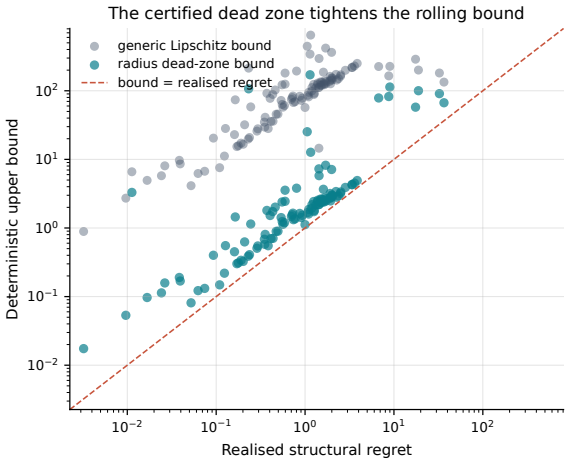
(a) Frozen lower and inverse upper brackets.



(b) Exact enumeration: candidate generation dominates.



(c) Exchangeable coverage and scale-shift controls.



(d) Generic and certified dead-zone bounds.

Figure 1: Audit summaries. The colours distinguish evidence strata rather than imply a common probability model.

At $T = 18$, the median instance generated 126,976 outside candidates; median bound generation took 12.10 s while the solved disjunct itself took 0.0051 s. Two of the six $T = 18$ exact-benchmark cases overran the 15 s wall limit during atomic candidate generation, before solving a disjunct; this leaves 28 of 30 benchmark instances optimal and two explicitly unresolved, with zero solver failures. A recorded hard case with 129,024 paths likewise overran a 5 s exact limit before solving a disjunct. Its polynomial bracket $[4.121, 6.128]$ took about 0.005 s, and a MILP probe at 1.01 times the safe lower bound proved "no" in about 0.56 s. Theorem 11.1 explains these observations exactly ($126{,}976 = 2^{17} - 2^{12}$ and $129{,}024 = 2^{17} - 2^{11}$), and the stratified evidence below explains the observed practicality boundary.

Corpus equivalence gate and preregistered scaling benchmark

Before any scaling claim, the compact model of section 11.4 was required to reproduce the release's own exact corpora (`results/raw/addendum_equivalence.csv`): the regenerated exact-certification instances ($T \leq 8$, all protected depths) and the exact-benchmark instances ($T \leq 18$). The 627 rows fall into five disjoint evidence classes. There are 305 `exact_match_certified` compact-model comparisons, each with an exactly re-verified witness; 247 `exact_rational_equality` comparisons between direct and decomposed certified enumeration. A further 55 benchmark-search rows form a platform-sensitive partition: the frozen macOS replay produced 27 `float_match_certified` comparisons whose numerical search result is paired with an exact witness check, and 28 `reference_unresolved` rows, compared with 30 and 25 respectively in the preliminary Linux replay. The remaining 20 rows are `unresolved_pair`. There were zero disagreements within the relation warranted by each class. The 48 unresolved rows remain part of the denominator and are not exact validations. The gate also exposed one defect in an early witness-repair path (a floating witness inflated by $1 + 2^{-12}$ before certification); certification now derives the exact binding-comparator tie instead.

The scaling benchmark was then run under a design frozen and hashed before execution (`docs/benchmark_preregistration_addendum.md`): $T \in \{18, 25, 35, 50, 75, 100\}$, families regular, near-tie and adversarial (floor-heavy and strict-gap), two replicates, protected depth $k = \lceil m/2 \rceil$ plus a full-path contrast at $T = 18$; per-solve caps of 60 s (180 s for enumeration) and a candidate-guard threshold of 2^{21}; 210 recorded solves, no cap or grid deviations during the run (one post-run summary recomputation from unchanged raw rows is recorded in the summary's deviation list). The preregistered outcomes:

- **P1 (largest resolved horizon).** Exhaustive enumeration resolved nothing beyond $T = 18$: all 30 attempts at $T \geq 25$ were refused by the closed-form candidate guard ($N_{\text{out}} > 2^{21}$, disclosed before generation, not a timeout). The arc-product decision model, both compact models and the polynomial bracket all reached $T = 100$; the compact minimum-radius model solved 36 of 42 instances to optimality (median 0.13 s at $T = 18$, 60 s at $T = 100$; six time-limit incumbents at the largest horizons), and every optimum carried an exactly re-verified witness whose radius respected the certified frozen lower bound in all 42 cases.
- **P2 (decision wall time).** On the 42 common probes the compact decision model was faster than the released arc-product model in every cell, with a median wall-time ratio of 28.7× (interquartile 16.4×–76.9×); it solved all 42 probes to optimality (median branch-and-bound node counts between 1 and 80), while the arc-product model hit its cap 13 times.
- **P3 (bracket width).** The polynomial certificate alone closed exactly in 32 of 42 cells (median relative width 0; maximum 0.646), at median 3.0 s even at $T = 100$ in exact rational arithmetic.

The practical conclusion matches theorem 11.2: the polynomial bracket plus the compact exact model is the deployable escalation path, and exhaustive enumeration is not an ordinary-horizon method — its refusals are now principled disclosures of a closed-form count rather than timeouts.

The tests were produced in the same workflow as the code. They increase internal confidence but do not constitute independent reproduction.

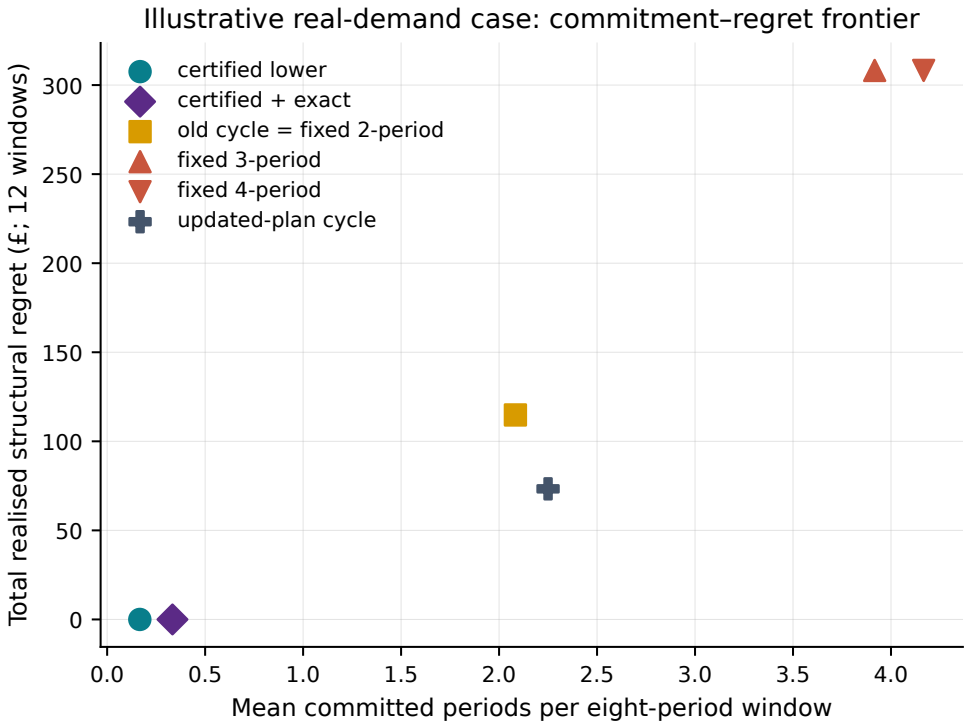


Figure 2: Illustrative commitment–regret frontier under the primary stylised cost design. Twelve windows are too few to support a comparative effectiveness claim.

15 Illustrative real-demand case

The UCI *Daily Demand Forecasting Orders* dataset contains 60 daily observations from a Brazilian logistics company, 12 predictors and a total-orders target, with no missing cells (Ferreira et al. 2016). The local CSV has SHA-256

`adf8dda021beae76b13b1cbb5b3c075d064f03b3353a711c1ddc357c6f89e924`.

The exact public-mirror URL used to obtain the local bytes was not preserved. The release therefore records a chain-of-custody gap, checks the file against the official UCI schema and hash-locks the analysed bytes. A clean reproduction should obtain the file from the official UCI archive, record its hash, and resolve any difference from the locked bytes before analysis.

This is *not* a field validation. Observed demand is real; forecast vintages and costs are reconstructed. A causal pooled direct ridge model creates 31 overlapping eight-day windows. The old forecast uses information through three days before the first target, the new forecast through two days before it. Only target lags, rolling target means and known target-calendar variables enter; contemporaneous future predictors from the source table are excluded. Eight windows estimate one pooled norm scale, 11 calibrate a split-conformal threshold at $\alpha = 0.20$, and 12 remain for retrospective policy evaluation. Reusing one threshold across all 12 evaluation windows has no simultaneous coverage guarantee.

Setup costs $\{\$60, \$120, \$240\}$ and daily holding costs $\{\$0.06, \$0.12, \$0.24\}$ per order are stylised: monetary symbols denote *stylised cost units parameterised in pounds*, not an empirical calibration of the organisation's economics. The primary design uses $K = \$120$ and $h = \$0.12$. The decision outcome is structural regret under realised demand in those stylised units, not the original organisation's financial loss. The 12 evaluation windows *overlap* and are not independent replicates; totals over them describe this one reconstruction only.

The case demonstrates four points. The software can ingest forecast vintages and return safe, unsafe or

unresolved boundaries. The lower certificate may be too conservative to commit much. Exact escalation can close a material boundary. Fixed fences trade more commitment for structural regret in this particular reconstruction. The sample, overlap, reconstructed vintages and stylised costs prevent generalisation.

16 Bounded decision protocol

The system returns a decision record, not a blanket instruction to commit.

Step 1. Freeze the object. Record $K, q, \widehat{d}, \ell, w$, units, forecast vintage, input hash and model exclusions. Verify ordered slopes.

Step 2. Audit nominal degeneracy. Construct the optimal-arc graph, report the longest common optimal prefix and use a deterministic tie-break for any complete comparator.

Step 3. Compute the certified bracket. Retain the exact-rational lower radius, exposed outside path, exact tie witness, any certified upper witness, the infinity status (no outside path, saturation, or not infinite) and search metadata for each k. Archive the decision record together with its independent-checker verification.

Step 4. Calibrate once for the stated decision. Align revision rows by product class, horizon, vintage and information set; audit missingness and drift; freeze weights before scoring. For repeated prespecified decisions, allocate error explicitly.

Step 5. Apply three-way triage. Commit only when the calibrated threshold is strictly below the certified lower radius. A threshold at or above a certified upper radius has an exactly verified destabilising witness. The interval between them is unresolved.

Step 6. Escalate selectively. Run the exact disjunction or fixed-radius MILP only when resolving the interval has enough decision value. Preserve all timeouts, gaps and lower/upper bounds. Never recode unresolved as safe.

Step 7. Monitor realised structural loss. Replan at disclosed cycle boundaries, maintain the information structure and quantity-recourse assumptions, and compare realised residuals with (33).

Step 8. Govern model exceptions. Capacity, minimum quantities, joint setups, supplier locks, service levels and manual overrides lie outside this certificate unless represented in the optimisation model.

17 Glossary and computational status vocabulary

18 Scope, failure modes and research frontier

Model scope. The theory covers one uncapacitated item, deterministic demand conditional on a scenario, linear costs, no backlogging and ordered reduced slopes. Capacity, joint setups, minimum order quantities, perishability, substitution and service constraints change the feasible family and may destroy the formulas.

Demand domain and semantic mode. The admissible set has coordinate floors and no extra upper bounds beyond the radius box. Forced-source graph mode permits zero first-period demand and costless empty setup nodes because those are explicit graph decisions. Classical active-horizon mode requires $\ell_1 > 0$ and $K_i > 0$ for $i \geq 2$; only under those conditions does theorem 2.2 transfer setup-prefix conclusions to the ordinary lot-sizing model. Directionally asymmetric norms require new pairwise and breakpoint derivations.

Nominal choice. The exact radius is a property of a node prefix. The frozen certificate is conditional on the selected complete nominal comparator. The release makes this dependency visible and recommends protecting only the common nominal prefix unless a disclosed secondary criterion selects among optima.

Numerical assurance. Every certified value is computed or a posteriori verified in exact rational arithmetic, decision records carry machine-checkable witnesses, and an independent checker re-verifies them; the previous version's tolerance-based safe margins are gone. The remaining assurance gaps are stated exactly: for $T > 12$ the checker re-verifies achievability, infinity and consistency but relies on the package's internally cross-checked parametric computation for lower-bound minimality; mixed-integer "no" decisions remain solver-level evidence unless the certified bracket already covers them; and neither implementation has proof-assistant verification. The exact layer decides the rational instance actually supplied: plain JSON numbers are read as their exact binary values, and users whose data are decimal should supply numeric strings, which are parsed as exact decimals.

Statistical scope. Exchangeability can fail under forecast-system changes, promotions, censoring, missing horizons or distribution shift. The negative controls show the direction of risk, not a universal correction. A sequential version should state its target, filtration and long-run or finite-horizon guarantee explicitly.

External validity. The synthetic audit tests correctness and boundary cases. The UCI case uses real demand but synthetic vintages and costs. A field study should preserve actual forecast vintages, release decisions, order quantities, capacity constraints, service outcomes and expediting costs, and compare against fixed fences and unrestricted replanning.

Open exact complexity. Generic partial inverse and partial anti-inverse shortest paths are already hard: NP-complete on DAGs even for one required arc with increase-only weights (Ley and Merkert 2025) and strongly NP-hard under L_1 and L_∞ for forbidden elements (Gassner 2009). The present analysis sharpens the boundary on both sides. Algorithmically, the released escalation is now proved worst-case exponential — it materialises $2^{T-1} - 2^{\max\{T-\nu_k,0\}}$ outside candidates before its first inverse programme (theorems 11.1 and 11.2) — while the decision problem itself lies in NP with polynomial-bit witnesses (theorem 2.5); an exponential implementation must not be mistaken for problem hardness (theorem 11.3). Structurally, the prefix radius decomposes exactly into at most k one-protected-arc radii (theorem 11.8), so the open question is now confined to its narrowest core: can a rational demand vector in the weighted floor-truncated box make one specified Wagner–Whitin outgoing arc cease to be uniquely optimal, under correlated costs $K_i + q_i(D_j - D_i)$ and ordered slopes? Neither a polynomial algorithm nor a hardness reduction preserving the demand-induced cost map is known. The compact production-balance model (theorem 11.12) supplies a polynomial-size exact formulation — solver work remains NP-type — and its linear relaxation is the natural place to search for integrality or a fractional counterexample seeding a reduction.

Novelty status. A page-specific reading of the supplied Richter–Vörös (1989) scan closes the source-specific question that motivated the acquisition: that paper studies scalar-parametric stability of one fixed regeneration-point solution and does not state the protected-prefix, moving-continuation or weighted demand-ball object developed here (section 3). The release records the inspected pages and the hash of the supplied Pages bundle. This is strong negative evidence against a collision in the closest identified source. It is not an exhaustive priority proof: the search is targeted, adjacent same-programme papers remain less completely inspected, and independent bibliographic attestation remains a later assurance target.

19 Conclusion

The decision object is the weighted distance from the current demand forecast to the nearest instance on which an optimal Wagner–Whitin path may leave a protected replenishment prefix. Reduced arcs $K_i + q_i(D_j - D_i)$ make this geometry tractable in layers.

A fixed outside path has an exact floor-aware tie radius. A formally specified prefix-deviation graph and a repaired Charnes–Cooper argument compute the frozen-comparator radius in polynomial time; the released implementation computes the same value in exact rational arithmetic through breakpoint binary search and finitely terminating parametric shortest paths, and ships every certificate with independently checkable witnesses. That value is a valid lower certificate, exact for a full path and sometimes strict for a partial prefix. A compact fixed-path inverse programme turns the exposed competitor into a feasible upper bound; a closing witness or coincident bounds can certify the exact strong radius without enumeration. The finite disjunction and mixed-integer model remain exact escalation mechanisms, but their practical scale is unresolved.

The escalation layer is now characterised without conflating algorithmic cost with problem complexity. The released disjunction is worst-case exponential by an explicit candidate count known before generation, and the implementation can refuse it before front-loading that work. The decision problem lies in NP under rational encoding; the strong radius is the minimum of at most k one-arc suffix radii; and every finite boundary has an optimal-face edge witness formed by one structural detour. A compact production-balance model gives a polynomial-size exact formulation, although solving it remains mixed-integer work. The exact complexity of the demand-induced one-arc problem remains open, and the paper states that frontier rather than inferring hardness from an exponential routine.

Nominal-degeneracy analysis prevents arbitrary solver ties from masquerading as commitment robustness. One-step conformal calibration gives a marginal probability statement only under exchangeability and domain alignment. The rolling theorem prices structural setup decisions with quantity recourse and supplies a deterministic dead zone. Producer-side tests and an illustrative real-demand case support implementation plausibility while leaving independent reproduction and field validity open.

The resulting contribution is a bounded decision architecture with two explicit semantic layers. Its graph theorems apply to the forced-source model; its classical setup-schedule interpretation applies only through the active-horizon bridge. Certified outputs are rational and witness-backed, and the interface distinguishes certified-safe, witnessed-unsafe, certified-interval, refused and solver-level states. The source-specific Richter–Vörös reading supports the contribution boundary while leaving exhaustive priority open. Capacity, multi-item coupling, service outcomes and live forecast processes remain outside the warranted claim and require new theory or field evidence.

Table 3: Theorem-level contribution boundary. "Inspected" distinguishes full text from abstract-level evidence.

Source	Object retained	Perturbation and result	Boundary relative to this paper
Richter and Vörös (1989a) (supplied seven-page print scan inspected 18 August 2026; hash-matched source-specific receipt)	One fixed regeneration-point Wagner–Whitin solution, stationary costs (s, h)	One scalar cost direction $(s + a\hat{s},\ h + a\hat{h})$, one scalar demand direction $\tilde{d} = d + b\hat{d}$, and the joint (a, b) region; stability intervals/regions from inequality systems over regeneration points; example and IBM-XT timings to $T = 15$	Direct antecedent: the inspected source contains no protected-prefix class, moving continuation, tie event, simultaneous weighted L_∞ ball, floors or complexity result. This is a source-specific negative reading, not exhaustive priority evidence.
Hoesel and Wagelmans (1993) (full text)	One optimal economic lot-sizing solution	Maximal individual parameter ranges, including one demand coordinate at a time	Does not provide the simultaneous weighted demand norm or allow the best continuation within a protected prefix to switch.
Chakravarti and Wagelmans (1998) (full text)	A fixed binary solution	Stability radius under independent objective-coefficient perturbations; polynomial when the underlying optimisation oracle is polynomial	The full-path case belongs to this fixed-solution tradition. Demand revisions here correlate many arc coefficients, and the protected object is a face of paths.
Zhang (2011); Li and Romero Morales (2013) (publisher record and full preprint)	One setup vector selected before interval-demand uncertainty	Minimax worst-case regret against the scenario-optimal plan; polynomial dynamic programmes, improved to $O(T^5)$ for demand-only uncertainty	Closest robust-complexity comparator, but the quantifier order and objective differ: no nearest-radius search and no crossing of moving inside/outside prefix envelopes.
Egri et al. (2014) (full repository manuscript)	Observed complete lot-size decisions across historical demand samples	Infer cost-parameter ratios by an inverse shortest-path linear programme	Direct inverse lot-sizing precedent, but it changes latent costs to rationalise observed solutions rather than demand to destabilise a protected prefix.
Richter and Weber (1994) (abstract/metadata inspected)	A solution as the terminal horizon grows	Sequential stability and sufficient conditions for monotonicity	Concerns horizon extension, not monotonicity in protected prefix depth under a fixed finite forecast.
This paper	Every optimum must retain a node prefix	Simultaneous floor-constrained weighted demand revisions; moving inside continuation; polynomial lower–upper bracket and exact escalation	The strongest defensible candidate novelty is the prefix-face geometry, semantic bridge, certificate/bracket, local arc decomposition and single-detour optimal-face witness, together with exact escalation and the compact model. The inspected 1989 source does not absorb these claims.

Table 4: Which result controls which arrow. Each guarantee is stated for its own estimand only.

Arrow	Controlling result	Nature of the guarantee
$F^{\text{old}} \to F^{\text{new}}$ (revision size)	Theorem 12.1	Probabilistic, marginal, one step, exchangeability required
$F^{\text{new}} \to$ plan structure	Theorems 7.1 and 8.1	Deterministic, exact arithmetic, conditional on the revision lying inside the certified radius
Commitment $\to$ cost under D^{real}	Theorem 13.3	Deterministic, pathwise worst case, quantity recourse assumed

Table 5: Producer-side audit. Counts are full denominators; exact, interval, floating-only and unresolved evidence are reported separately.

Claim or component	Audit denominator	Result
Classical semantic bridge	320 exact active-horizon instances, $T = 1, \ldots, 10$	Graph and independent physical enumeration agreed on optimal cost and the complete set of optimal setup patterns in 320/320 cases; no optimal graph path contained an empty later setup. A graph-only negative control reproduced cost 101 versus classical cost 1.
Optimal-face edge witness	315 prefix cases from 84 exact instances, $T = 2, \ldots, 8$	All 254 finite exact boundaries exposed an adjacent inside/outside optimal pair with one divergence–remergence detour; the other 61 cases were proven infinite in the audited domain; no unresolved case or witness failure.
Frozen polynomial certificate	512 comparisons from 258 generated instances	Maximum absolute discrepancy from exhaustive frozen enumeration: 0.
Prefix-deviation graph	62 exhaustive small graphs	Every expanded/physical path set matched; every graph was acyclic.
Fixed outside-path inverse LP	96 comparisons	Compact suffix-potential LP matched an explicit inside-path LP to 4.55×10^{-13}; these are floating search checks followed by exact witness verification where certification is claimed.
Polynomial bracket	512 cases	411 exact closures and 101 certified intervals; among 254 partial-prefix cases, 153 closed exactly. Full-path cases closed as predicted by the theorem.
Exact certification	330 brute-force comparisons	330/330 exact frozen-certificate matches; 305 strong radii point-certified and 25 returned witness-backed certified intervals; maximum relative interval width 9.10×10^{-13}.
Compact-model equivalence	627 rows in five disjoint evidence classes	305 `exact_match_certified`; 247 `exact_rational_equality`; a platform-sensitive 55-row search partition (27 `float_match_certified` and 28 `reference_unresolved` in the frozen macOS replay); 20 `unresolved_pair`; zero warranted disagreements.
Exact enumeration benchmark	30 instances with a 15 s limit including candidate generation	28 optimal; two $T = 18$ cases timed out during atomic candidate generation before processing any disjunct; zero solver failures.
Fixed-radius MILP	24 tolerance-separated probes	22 optimal/resolved and correct; two $T = 16$ probes ended with feasible incumbents at the 2 s limit and remained unresolved.
Calibration	Four 20,000-trial scenarios and one 5,000-family scenario	Exchangeable coverage 0.9038 (Wilson 95% interval 0.8996–0.9078) for target 0.90; scale-shift controls fell to 0.8409 and 0.7791; Bonferroni family coverage 0.8984. These are Monte Carlo estimates, not theorem verification.
Rolling bound	462 cases, including zero and infinite radii	No mathematical violation; maximum output-level floating excess 2.28×10^{-13}; all 254 cases inside the certified dead zone had zero residual.

Table 6: Certified bracket runtimes (exact rational arithmetic, single core, worst case over two seeds and $k \in \{1, 3\}$).

T	Deviation arcs	Shortest paths	Newton steps	Median frozen (s)	Max bracket (s)
50	1,360	≤ 10	≤ 3	0.53	0.68
100	5,233	≤ 12	≤ 4	4.53	5.33
200	20,490	≤ 13	≤ 4	46.59	61.97
365	67,510	≤ 16	≤ 6	443.40	586.27

Table 7: Primary illustrative policy comparison over 12 reconstructed, overlapping evaluation windows. Regret is in stylised cost units parameterised in pounds.

Policy	Mean periods committed	Total structural regret (stylised)	Boundary record
Polynomial certificate only	0.17	£0.00	One unresolved boundary.
Certificate plus exact escalation	0.33	£0.00	One escalation; no unresolved boundary.
Old-plan one cycle / fixed 2-period fence	2.08	£114.80	Revision retained the commitment in 10 of 12 windows.
Fixed 3-period fence	3.92	£308.22	Revision retained the commitment in 10 of 12 windows.
Fixed 4-period fence	4.17	£308.22	Regret coincides with the 3-period fence here: the additional committed period matched the unrestricted benchmark in every window.
Updated-plan one cycle	2.25	£73.33	Formed after the later reconstructed vintage.
Unrestricted realised-demand benchmark	0	£0.00	Clairvoyant and not implementable.

Table 8: Outputs and their bounded meanings.

Output	Warranted statement	Unwarranted inference
Certified ρ^{lo}	Every admissible revision strictly inside it retains the prefix; computed in exact arithmetic	It need not equal the exact strong radius.
Certified ρ^{up}	An exactly verified destabilising witness exists at that radius	It does not prove no closer outside path exists.
Proven infinity	Either no outside path exists, or a certified positive residual gap survives maximal admissible reduction	Solver-presumed infinities are never published as proven.
Closed bracket / exact result	Nearest structural prefix change under the stated model	It is not a forecast probability.
Conformal threshold	One-step marginal revision coverage under exchangeable scores	It is not conditional or time-uniform under drift.
Rolling bound	Worst-case structural regret with quantity recourse	It is not expected financial loss or service impact.
Illustrative case	End-to-end feasibility on real observed demand with reconstructed inputs	It is not field validation.

Table 9: Terms used in the theorem statements, software records and audit files.

Term or status	Meaning
Forced-source graph	The declared path model in which every complete path starts at period 1 and path nodes remain distinct even when an empty zero-cost setup separates them.
Classical active horizon	The ordinary uncapacitated lot-sizing interpretation under $\ell_1 > 0$ and $K_i > 0$ for $i \geq 2$, as transferred by theorem 2.2.
Inside envelope A_k	Minimum cost over all continuations that retain the protected node prefix. Its optimiser may change with demand.
Outside envelope B_k	Minimum cost over paths that leave the protected prefix at or before its end.
Strong radius $\rho_{k,w}$	Distance to the nearest admissible demand at which an outside optimum ties or beats the inside envelope.
Frozen radius $\rho^{\mathrm{fr}}_{k,w}$	Polynomial lower certificate obtained by comparing every outside path with one fixed nominal complete path.
Single-detour edge witness	A co-optimal inside/outside adjacent path pair that differs only between one divergence and one remergence node.
`exact_certified`	Candidate enumeration and exact verification close the strong radius to one rational value.
`certified_interval`	Exact lower and feasible upper witnesses are valid, but the nearest value is not closed.
`enumeration_completed_with_certified_interval`	All permitted candidates were processed, but at least one disjunct remains unresolved; completion of enumeration is not exact certification.
`refused_by_candidate_guard`	The closed-form candidate count exceeded the predeclared cap; no safety conclusion follows.
`solver_level_only`	A mixed-integer search result lacks the exact witness or closing bound needed for a certified classification.

A Proof details and edge cases for the polynomial certificate

For an outside path Q, write $c_{Q,t} = \beta_{Q,t} - \beta_{P^\star,t}$. Its worst frozen gap at radius r is

$$F_Q(r) = \kappa_Q - \kappa_{P^\star} + \sum_{c_{Q,t}<0} c_{Q,t}(\widehat{d}_t + r/w_t) + \sum_{c_{Q,t}>0} c_{Q,t} \max\{\ell_t, \widehat{d}_t - r/w_t\}.$$

The expanded graph adds the same setup cost and coefficient contribution once for every physical arc, so (12)–(14) reproduce F_Q exactly.

If distinct breakpoints coincide numerically, the implementation collapses them before forming intervals. A coordinate with $\widehat{d}_t = \ell_t$ is saturated at radius zero. Both neighbouring formulae agree at a finite endpoint, so assigning a tie to the first interval does not change the earliest crossing. If $(b^\ell)^\top x = 0$, then F_x is constant on the interval and cannot cross after a safe left endpoint. If every path has zero denominator on the final interval, all remaining positive differences are floor-saturated and no negative difference is available to reduce a gap; the radius is infinite.

The linear programme can return a convex combination when several paths share the same ratio. Since a linear objective attains its optimum at an extreme point, one may recover a path by resolving with an infinitesimal deterministic arc-order tie-break or by running a shortest path at the returned boundary. Substitution into (10) gives a directly checkable half-space witness. The implementation checks that the recovered path's pairwise radius lies in the active interval; otherwise it reports that numerical witness recovery failed rather than silently certifying a path.

B Exact destabilising set as a union of polyhedra

For each $Q \in O_k$, define

$$\mathcal{D}_Q = \{d \geq \ell : C_Q(d) \leq C_P(d) \text{ for every } P \in \mathcal{I}_k\}.$$

Then

$$\mathcal{D}_k = \bigcup_{Q \in O_k} \mathcal{D}_Q, \qquad \rho_{k,w} = \mathrm{dist}_{\infty,w}(\widehat{d}, \mathcal{D}_k).$$

Each $\mathcal{D}_Q$ is a polyhedron. It has an exponential direct description with one inequality per inside path and the compact extended description in (19). The frozen certificate replaces $\mathcal{D}_Q$ by the larger half-space $\{d : C_Q(d) \leq C_{P^\star}(d)\}$, explaining both safety and possible conservatism.

C Computational design, exclusions and missingness

Table 10: Frozen computational design.

Component	Design and missingness treatment
Certificate audit	258 generated instances: 162 broad synthetic and 96 conditioned adversarial. Every generated instance appears in the raw manifest. The conditioned set targets strictness and is excluded from prevalence interpretation. Partial prefixes and full paths are both tested.
Graph audit	Exhaustive path-set comparison for 62 small expanded graphs. No missing or failed graph cases.
Fixed competitor	96 compact LPs compared with separately constructed explicit inside-path LPs. No solver failures.

Component	Design and missingness treatment
Exact benchmark	30 instances across five horizons, three regimes and two replicates per cell. Exact limit 15 s includes candidate generation. Twenty-eight were optimal; two $T = 18$ cases timed out during atomic candidate generation before processing a disjunct. No solver failure occurred.
MILP benchmark	24 fixed-radius probes with 2 s limits. Two returned feasible incumbents without closing bounds and remain unresolved; they are not counted as correct negative or positive decisions.
Calibration	Trial counts, calibration sizes, ranks, covered counts and Wilson intervals appear in raw and processed files. Scale-shift rows are negative controls to which the exchangeability theorem does not apply. Inadmissible floor revisions receive infinite scores rather than being dropped.
Rolling	240 broad cases, 120 strict-gap conditioned cases, 100 exact-witness destabilising cases and two edge cases. All realised vectors satisfy floors. Frequencies in conditioned strata are not prevalence estimates.
UCI data	60 rows, 13 columns and zero missing cells in the analysed CSV. The exact local mirror URL is missing from provenance; official UCI metadata, schema, licence and DOI are recorded, and the bytes are hash-locked. Forecast vintages and costs are reconstructed and clearly separated from observed demand.
Software	Python 3.12.11, NumPy 2.3.5, SciPy 1.17.0 with embedded HiGHS 1.8.0 (MIT), pandas 2.2.3 and Matplotlib 3.10.8 in the frozen macOS run. HiGHS proposes LP/MIP search results but has no certification role. The release includes a lockfile, one-command verifier and deterministic seed. A producer-created fresh environment replay passed with isolated wheel import; no independent clean-room environment has replayed the package.

D Notation

Symbol	Meaning
T	Number of periods.
K_i, p_i, h_i	Setup, production/purchase and one-period holding costs.
H_i, q_i	Cumulative holding adjustment and ordered reduced slope.
D_j	Demand accumulated before node j.
$P^\star, \pi_k$	Selected nominal complete path and its first k arcs.
$\mathcal{I}_k, O_k$	Paths retaining and leaving the protected prefix.
A_k, B_k	Best inside and outside path-cost envelopes.
w, ℓ	Positive norm weights and coordinate demand floor.
$\rho_{k,w}$	Exact strong prefix-stability radius.
ρ^{fr}	Frozen-comparator lower radius.
$\rho^{\mathrm{lo}}, \rho^{\mathrm{up}}$	Polynomial lower and feasible upper bracket.
κ_P, β_P	Constant and demand-coefficient vector of path P.
Γ_a	Exact weighted coefficient diameter of suffix paths.
$R_\pi(d)$	Bellman residual of an executed structural block.
$V_i(d)$	Optimal suffix value from node i; $V_{T+1} = 0$.
E_r, O_r	Protected-arc and best-diverging values at v_r (section 11.2).
ρ_r	One-protected-arc radius at (v_r, v_{r+1}).
$N(i), \lvert O_k \rvert$	Suffix path count $2^{\max\{T-i,0\}}$ and outside-path count.

y, x, I	Setup, production and inventory variables of the compact model.

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